

CSE 385: Analysis of Algorithms (Honors)

Lecture 6 (Divide-and-Conquer Algorithms: Matrix Multiplication)

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Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

$$\begin{bmatrix} z_{11} & z_{12} & \cdots & z_{1n} \\ z_{21} & z_{22} & \cdots & z_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ z_{n1} & z_{n2} & \cdots & z_{nn} \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{bmatrix} \times \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1n} \\ y_{21} & y_{22} & \cdots & y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nn} \end{bmatrix}$$

Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

Z	1	2	3	4
1				
2				
3				
4				

=

X	1	2	3	4
1	1	0	5	1
2	3	2	0	9
3	8	1	6	0
4	0	3	1	4

×

Y	1	2	3	4
1	3	3	0	2
2	0	1	5	3
3	7	0	2	1
4	1	1	5	2

Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

Y	1	2	3	4
1	3	3	0	2
2	0	1	5	3
3	7	0	2	1
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X	1	2	3	4
1	1	0	5	1
2	3	2	0	9
3	8	1	6	0
4	0	3	1	4

	1	2	3	4
1				
2				
3				
4				

Z

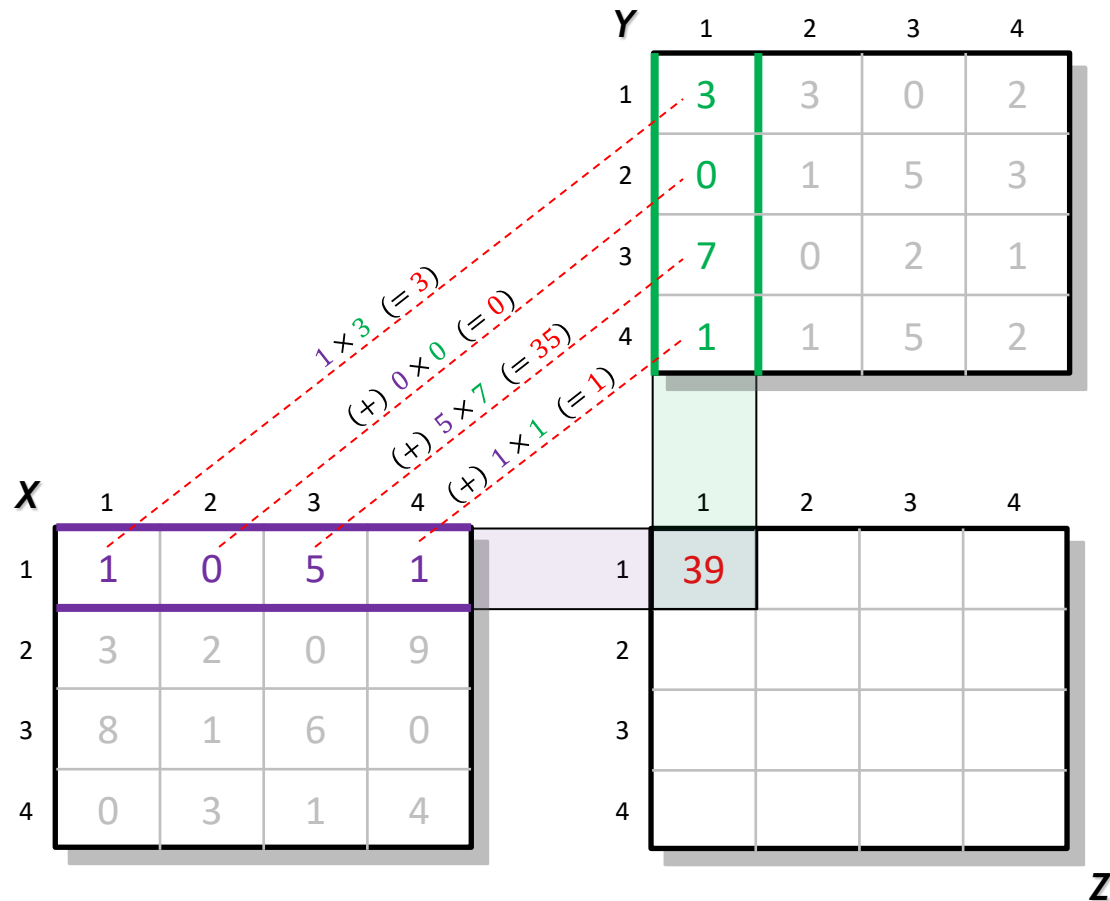
Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

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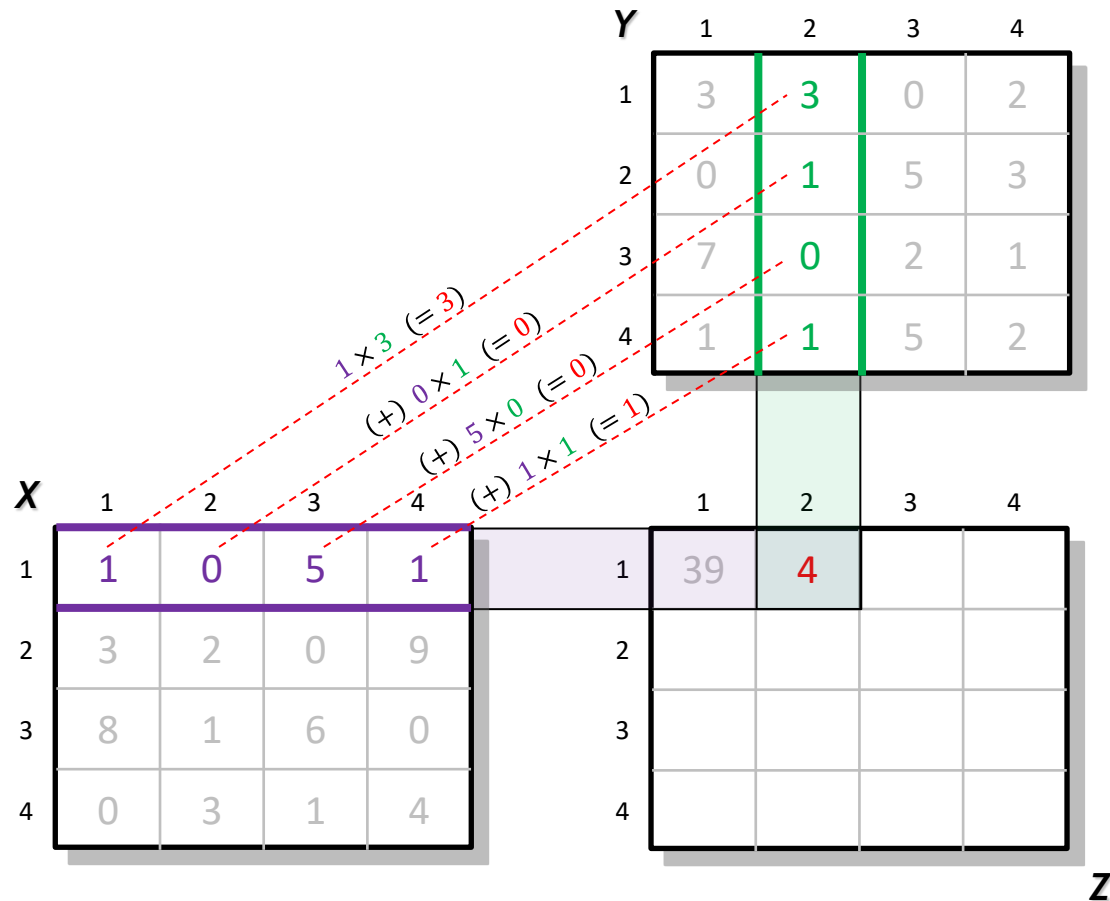
Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$



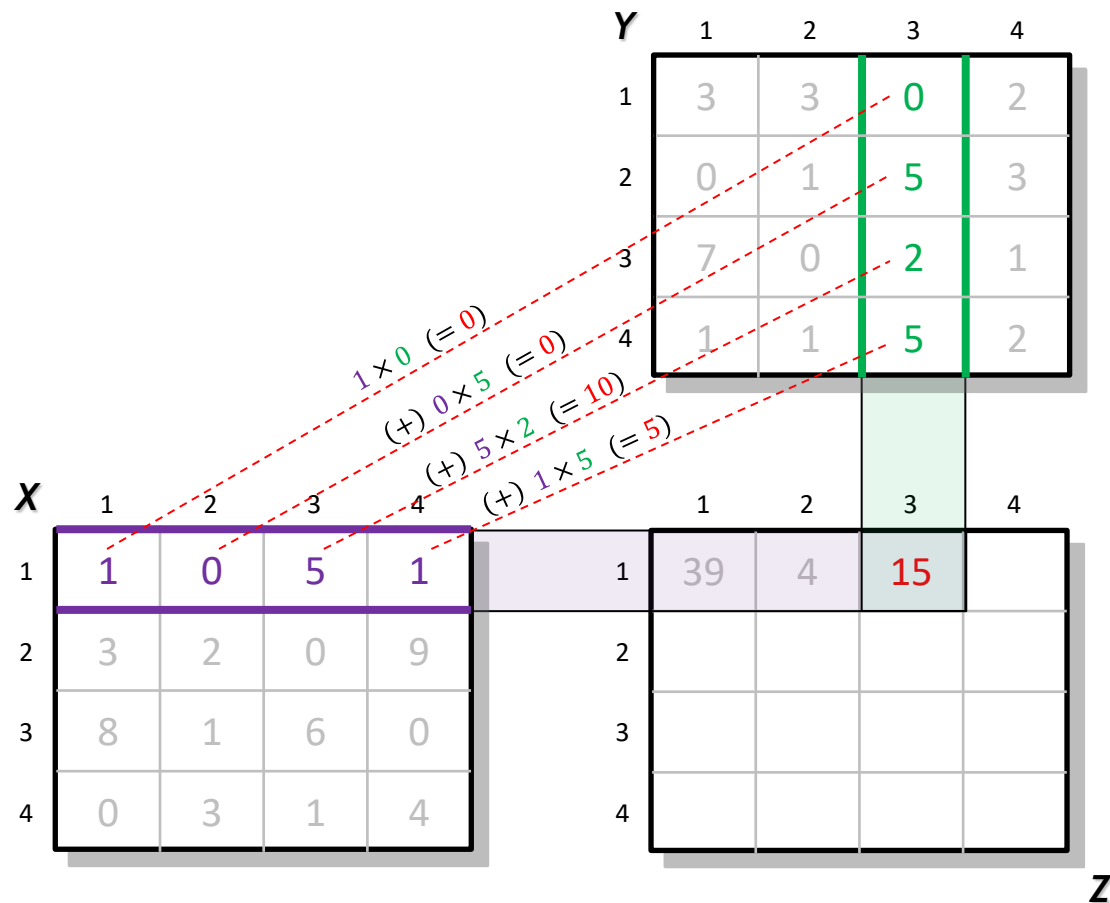
Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$



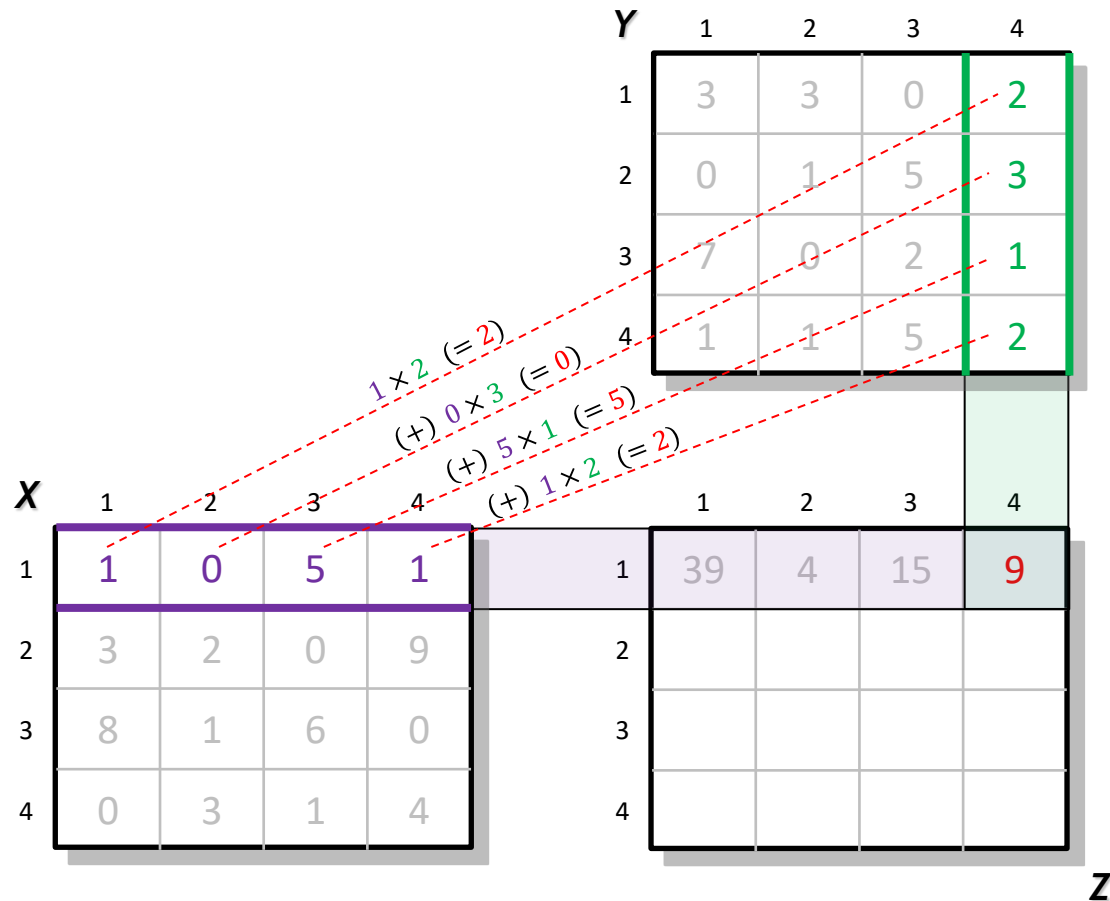
Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$



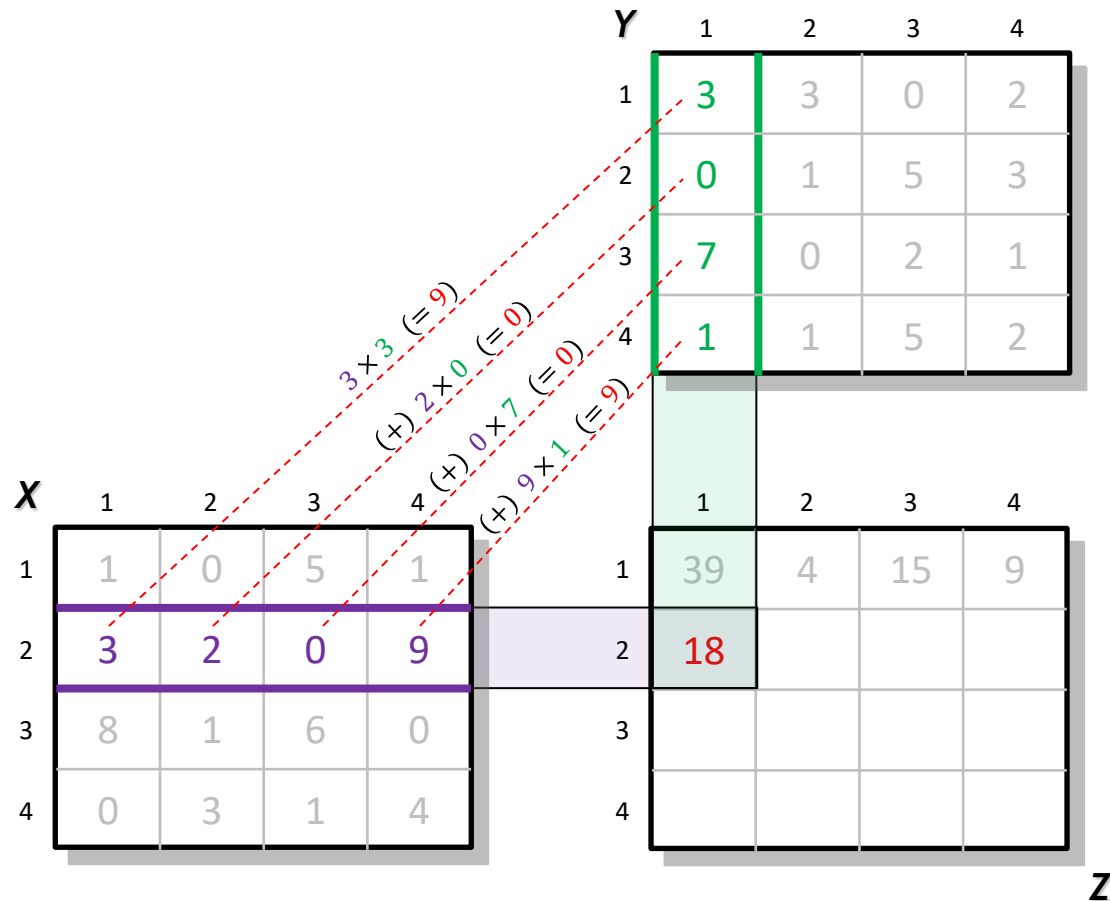
Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$



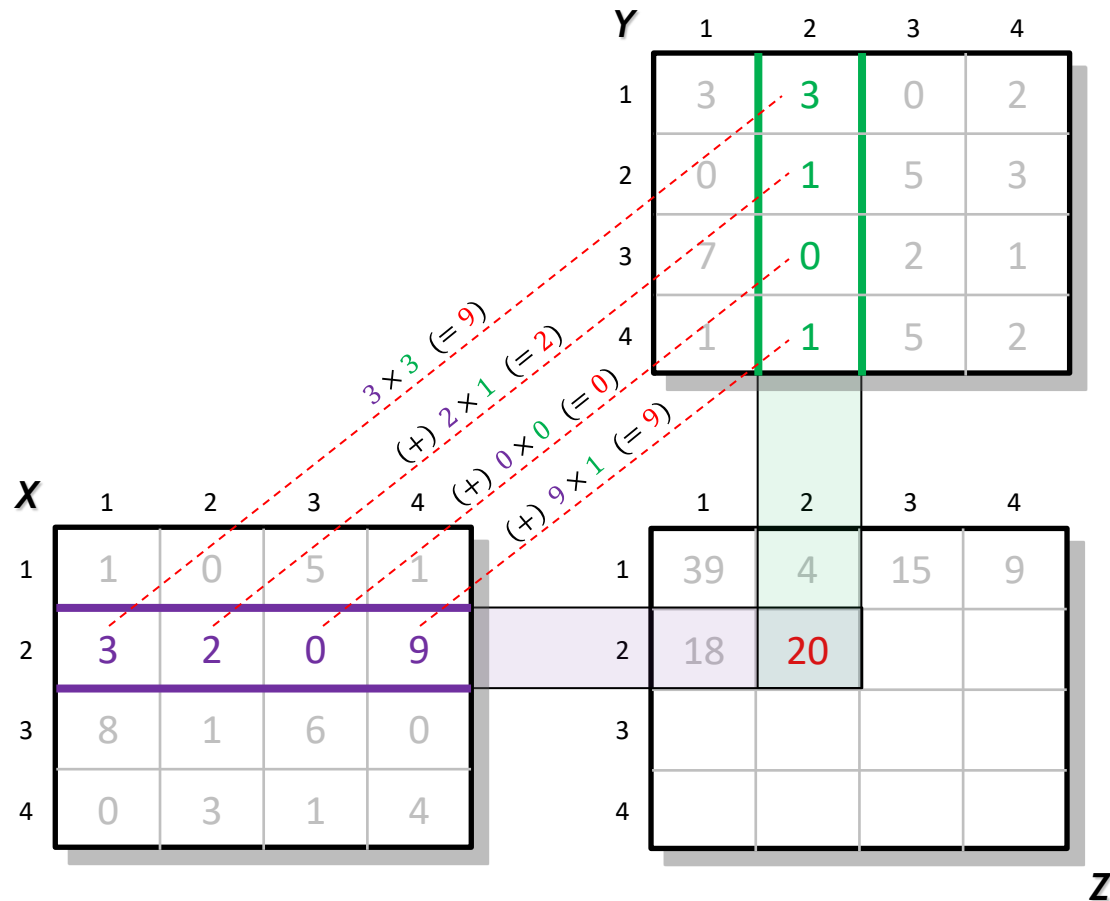
Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$



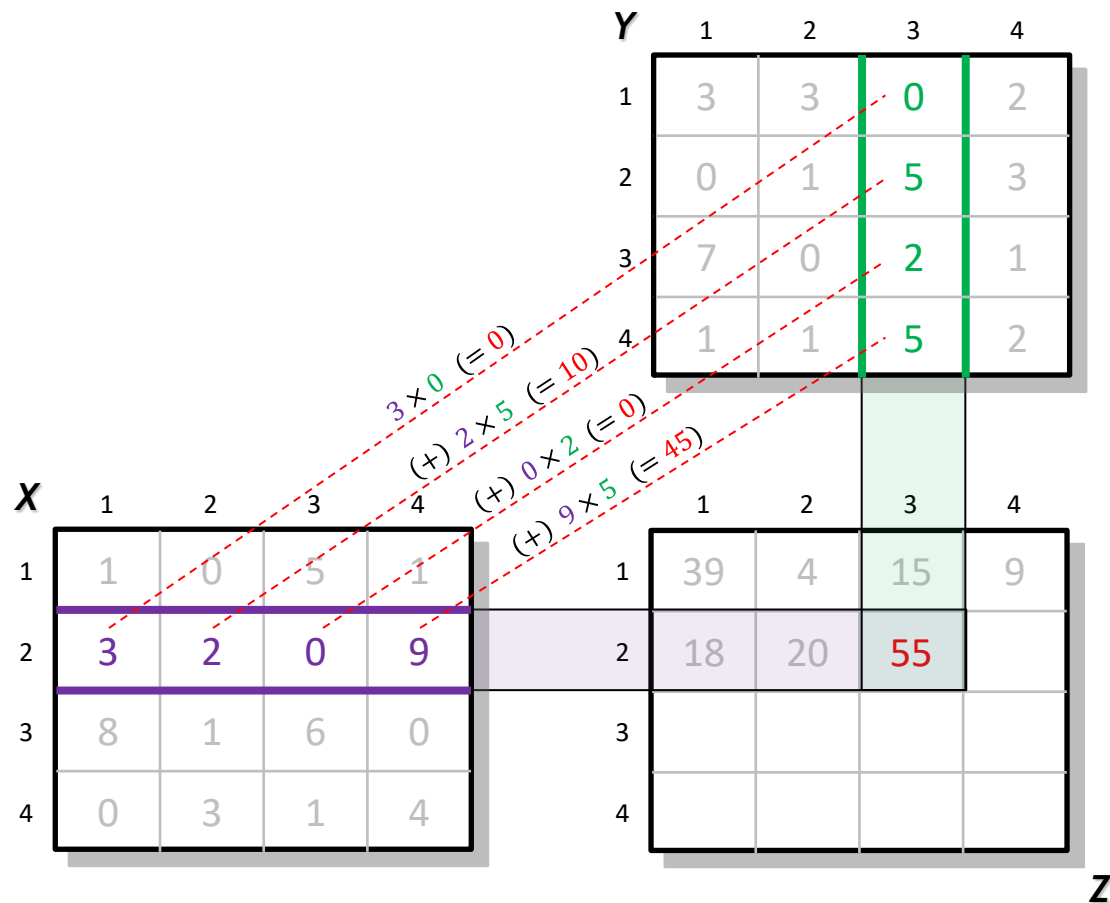
Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$



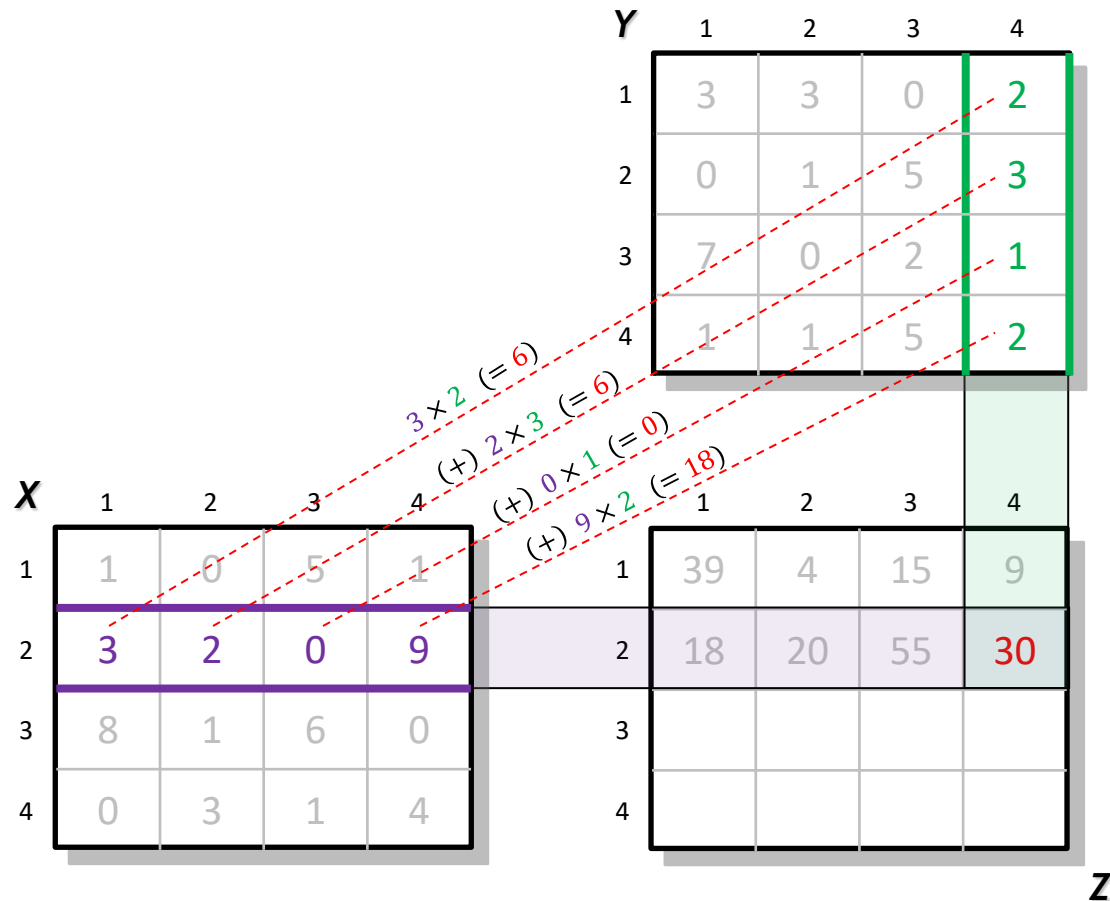
Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$



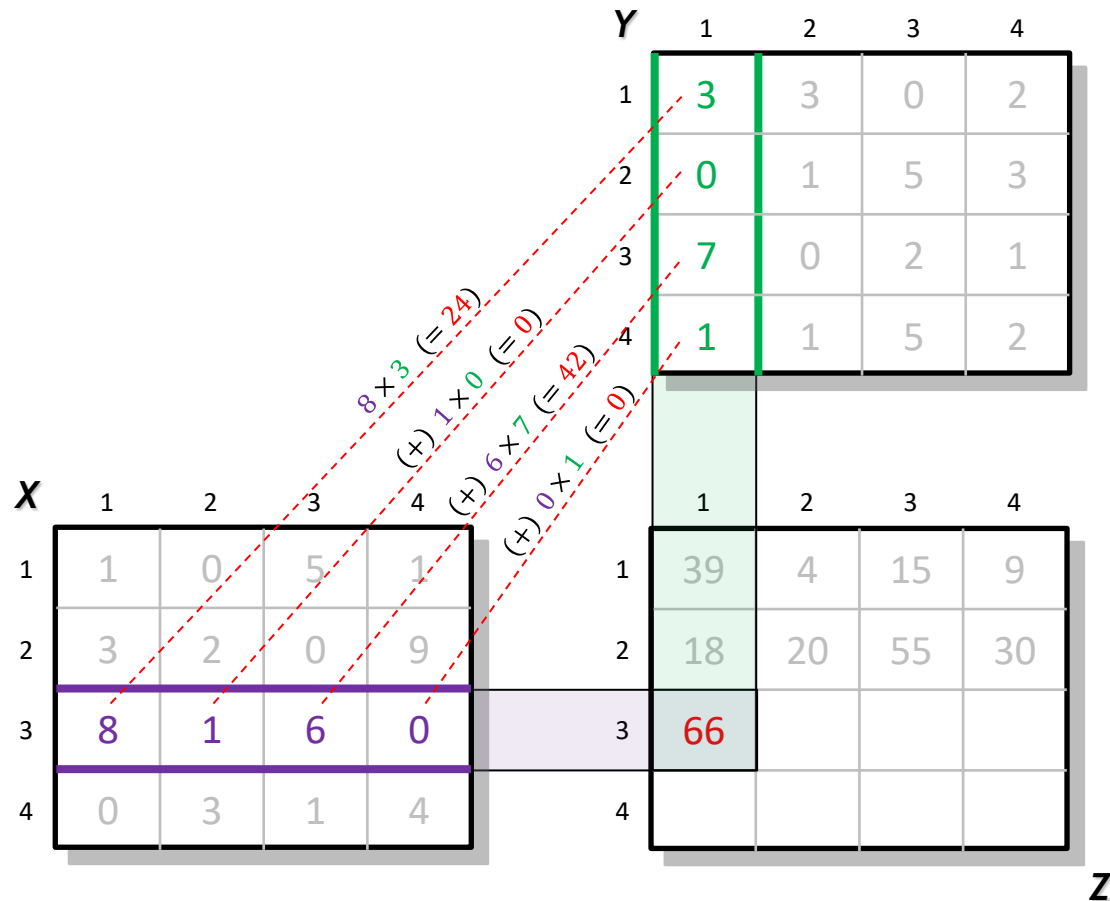
Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$



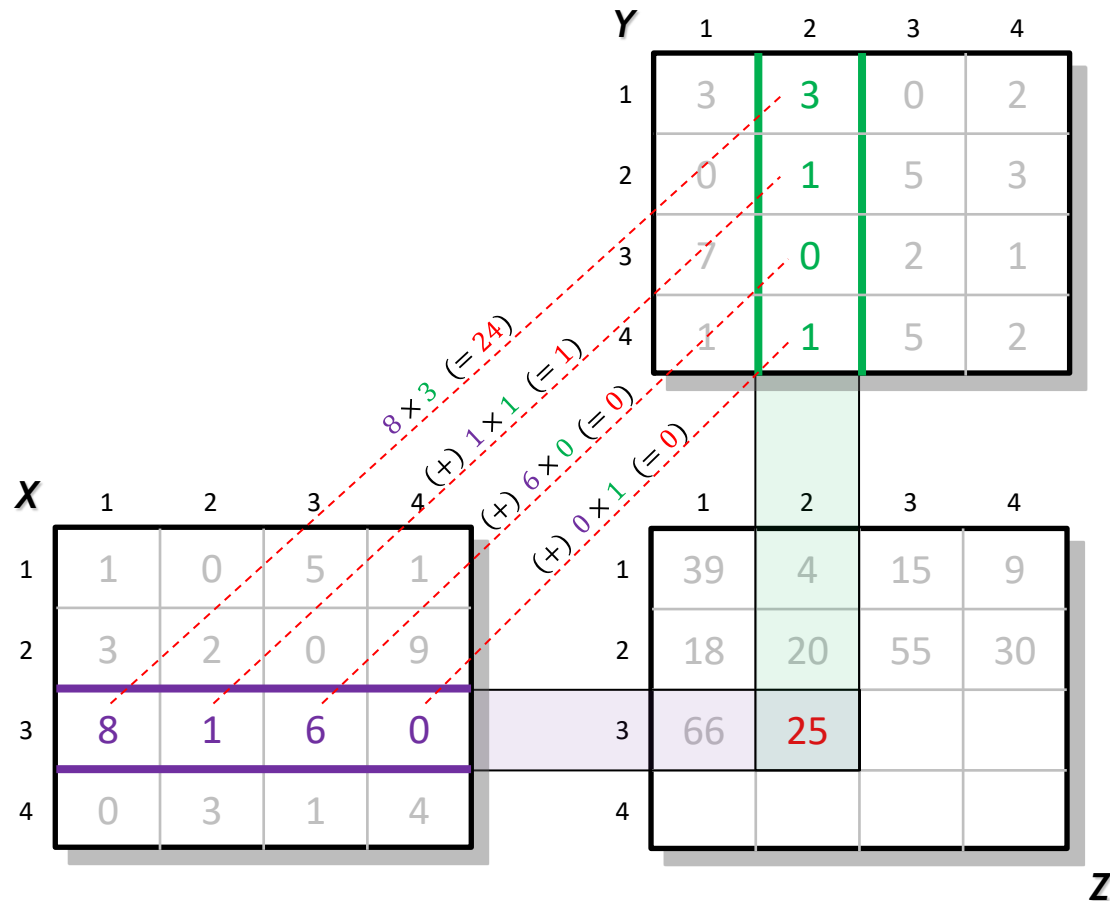
Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$



Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$



Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

		Y						
		1	2	3	4			
X	1	3	3	0	2			
	2	0	1	5	3			
	3	7	0	2	1			
	4	1	1	5	2			
		1	2	3	4			
1	1	0	5	1	39	4	15	9
2	3	2	0	9	18	20	55	30
3	8	1	6	0	66	25	17	
4	0	3	1	4				

Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

		Y			
		1	2	3	4
X	1	3	3	0	2
	2	0	1	5	3
	3	7	0	2	1
	4	1	1	5	2
		1	2	3	4
1	1	0	5	1	
2	3	2	0	9	
3	8	1	6	0	
4	0	3	1	4	
		1	2	3	4
1	39	4	15	9	
2	18	20	55	30	
3	66	25	17	25	
4					

Z

Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

		Y			
		1	2	3	4
X	1	3	3	0	2
	2	0	1	5	3
	3	7	0	2	1
	4	1	1	5	2
		1	2	3	4
Z	1	39	4	15	9
	2	18	20	55	30
	3	66	25	17	25
	4	11			

Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

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Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

The figure displays three 4x4 grids, labeled X, Y, and Z, arranged in a triangular pattern. Each grid contains numerical data, with specific rows and columns highlighted in different colors to indicate relationships or operations.

Grid X (Bottom Left):

	1	2	3	4
1	1	0	5	1
2	3	2	0	9
3	8	1	6	0
4	0	3	1	4

Grid Y (Top Right):

Y	1	2	3	4
1	3	3	0	2
2	0	1	5	3
3	7	0	2	1
4	1	1	5	2

Grid Z (Bottom Right):

	1	2	3	4
1	39	4	15	9
2	18	20	55	30
3	66	25	17	25
4	11	7	37	

Connections and Highlights:

- A vertical line connects the third column of Grid Y to the third column of Grid Z.
- A horizontal line connects the bottom row of Grid X to the bottom row of Grid Z.
- The bottom row of Grid X and the third column of Grid Z are highlighted in purple.
- The third and fourth columns of Grid Y are highlighted in green.

Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

Y

1

2

3

4

1

2

3

4

3

3

0

2

0

1

5

3

7

0

2

1

1

1

5

2

X

1

2

3

4

1

2

3

4

1

0

5

1

3

2

0

9

8

1

6

0

0

3

1

4

Z

1

2

3

4

1

2

3

4

39

4

15

9

18

20

55

30

66

25

17

25

11

7

37

18

Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

Y	1	2	3	4
1	3	3	0	2
2	0	1	5	3
3	7	0	2	1
4	1	1	5	2

X	1	2	3	4
1	1	0	5	1
2	3	2	0	9
3	8	1	6	0
4	0	3	1	4

	1	2	3	4
1	39	4	15	9
2	18	20	55	30
3	66	25	17	25
4	11	7	37	18

Z

Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

Z	1	2	3	4
1	39	4	15	9
2	18	20	55	30
3	66	25	17	25
4	11	7	37	18

=

X	1	2	3	4
1	1	0	5	1
2	3	2	0	9
3	8	1	6	0
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×

Y	1	2	3	4
1	3	3	0	2
2	0	1	5	3
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Iterative Matrix Multiplication

$$z_{ij} = \sum_{k=1}^n x_{ik} y_{kj}$$

$$\begin{bmatrix} z_{11} & z_{12} & \cdots & z_{1n} \\ z_{21} & z_{22} & \cdots & z_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ z_{n1} & z_{n2} & \cdots & z_{nn} \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{bmatrix} \times \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1n} \\ y_{21} & y_{22} & \cdots & y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nn} \end{bmatrix}$$

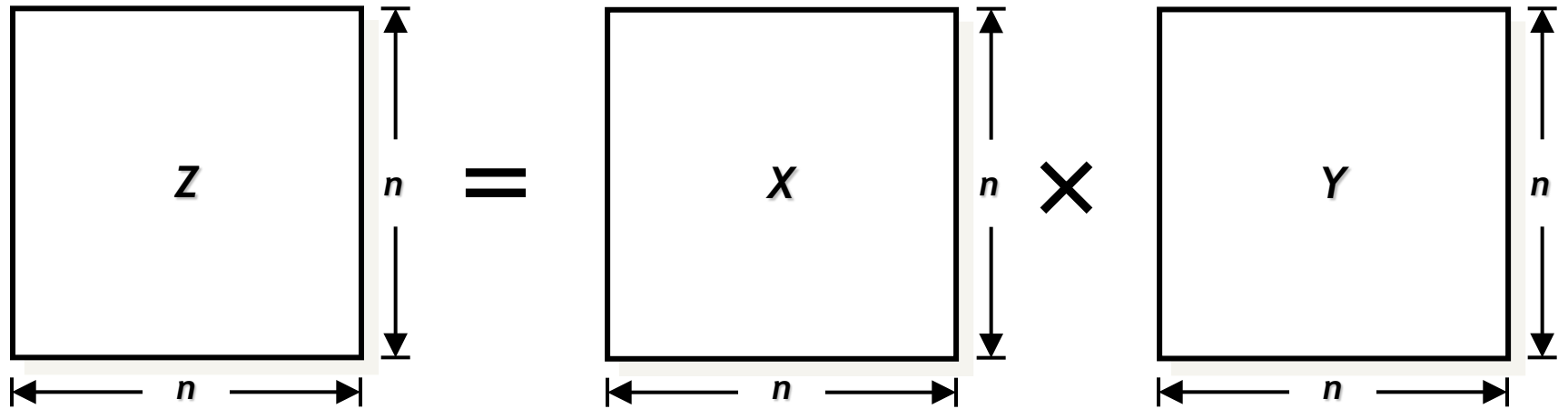
Iter-MM (Z, X, Y)

{ X, Y, Z are $n \times n$ matrices,
where n is a positive integer }

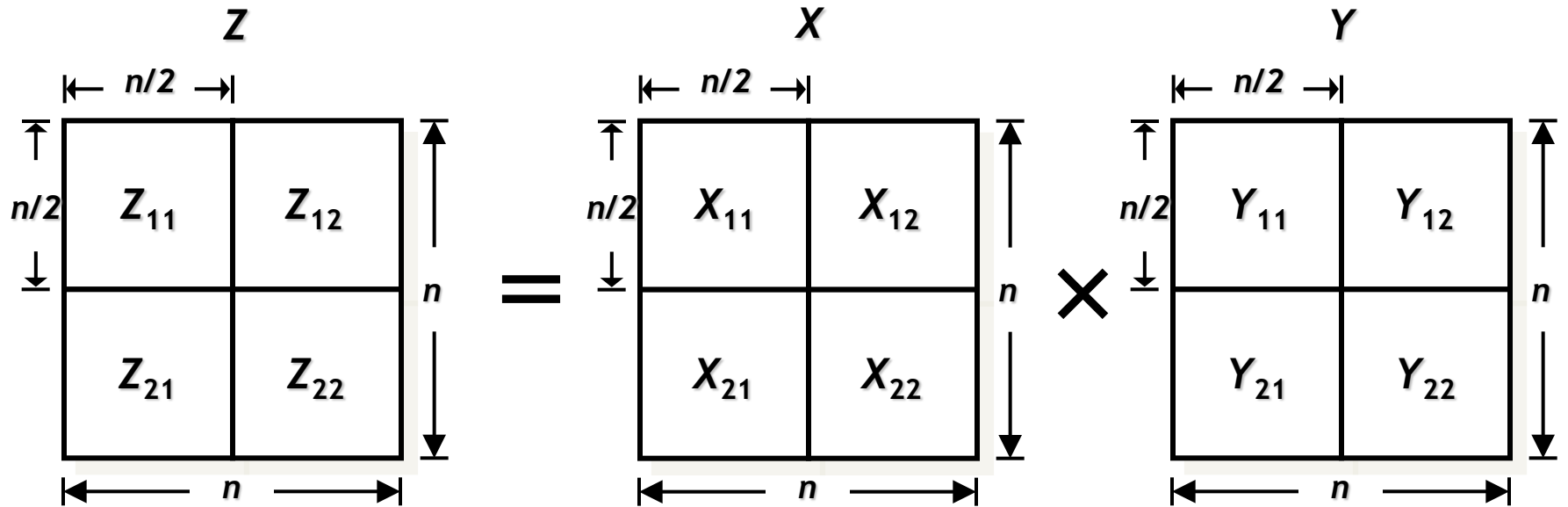
1. *for* $i \leftarrow 1$ *to* n *do*
2. *for* $j \leftarrow 1$ *to* n *do*
3. $Z[i][j] \leftarrow 0$
4. *for* $k \leftarrow 1$ *to* n *do*
5. $Z[i][j] \leftarrow Z[i][j] + X[i][k] \cdot Y[k][j]$

Running time, $T(n) = \Theta(n^3)$

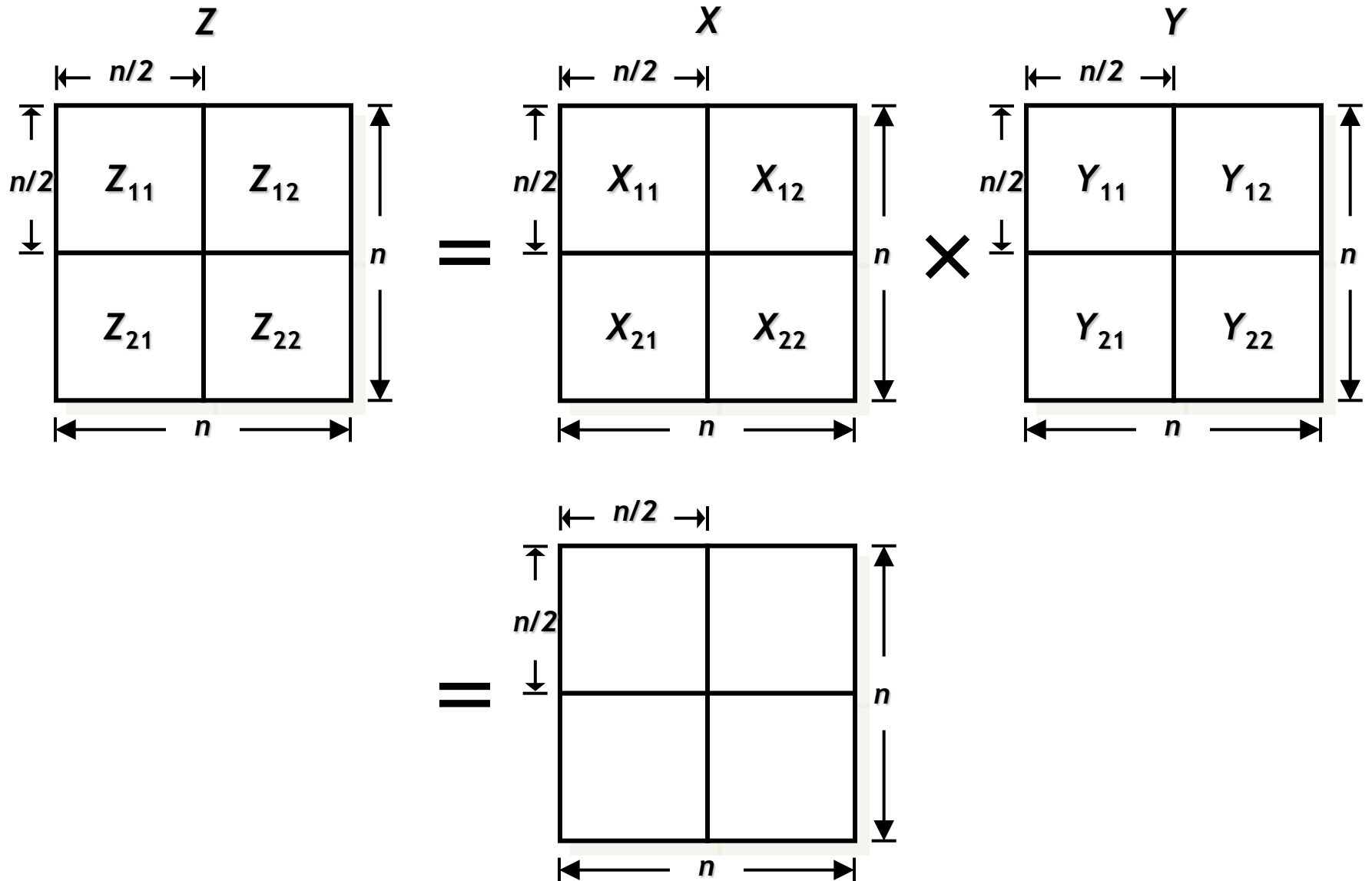
Recursive (Divide & Conquer) Matrix Multiplication



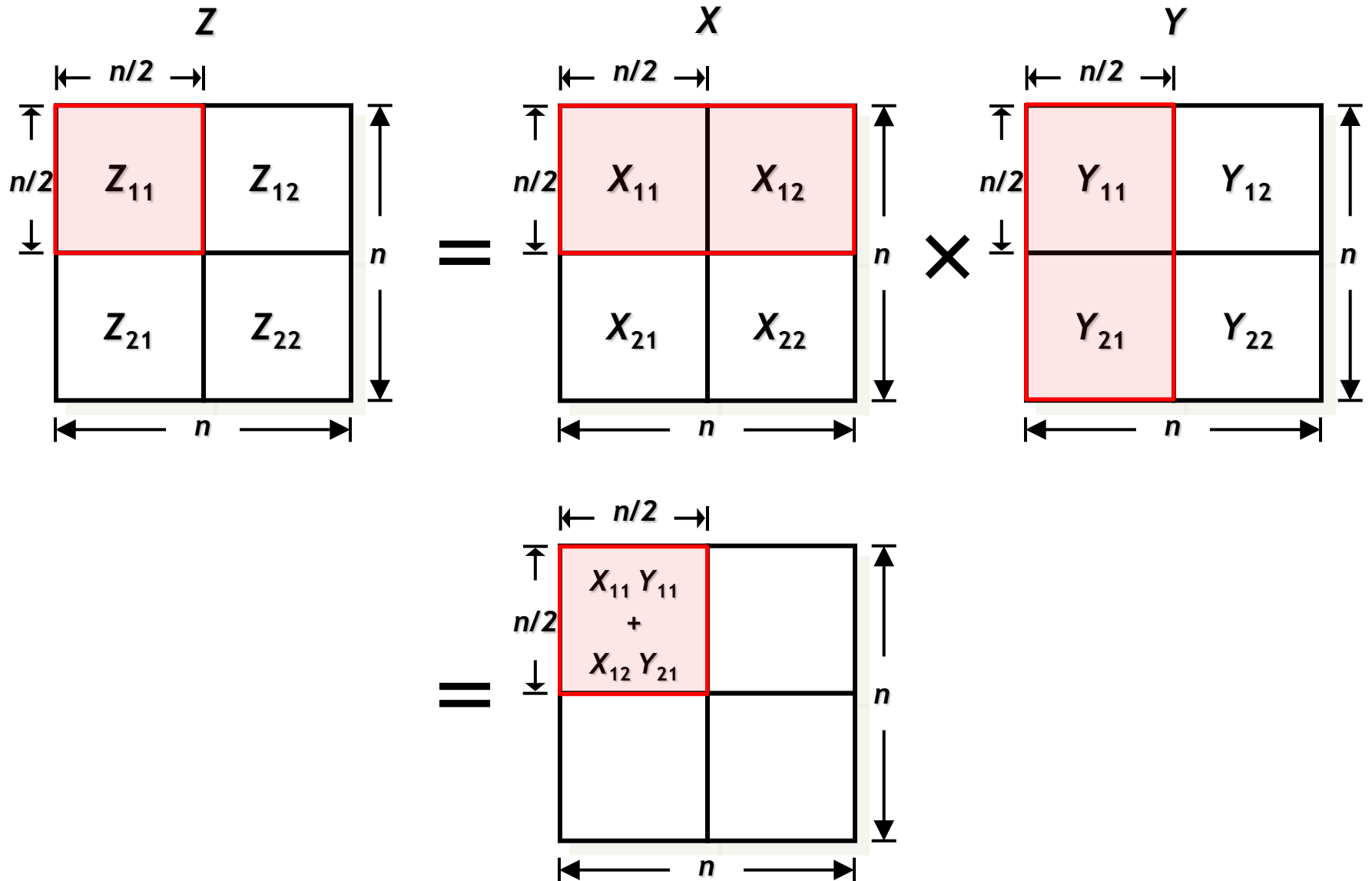
Recursive (Divide & Conquer) Matrix Multiplication



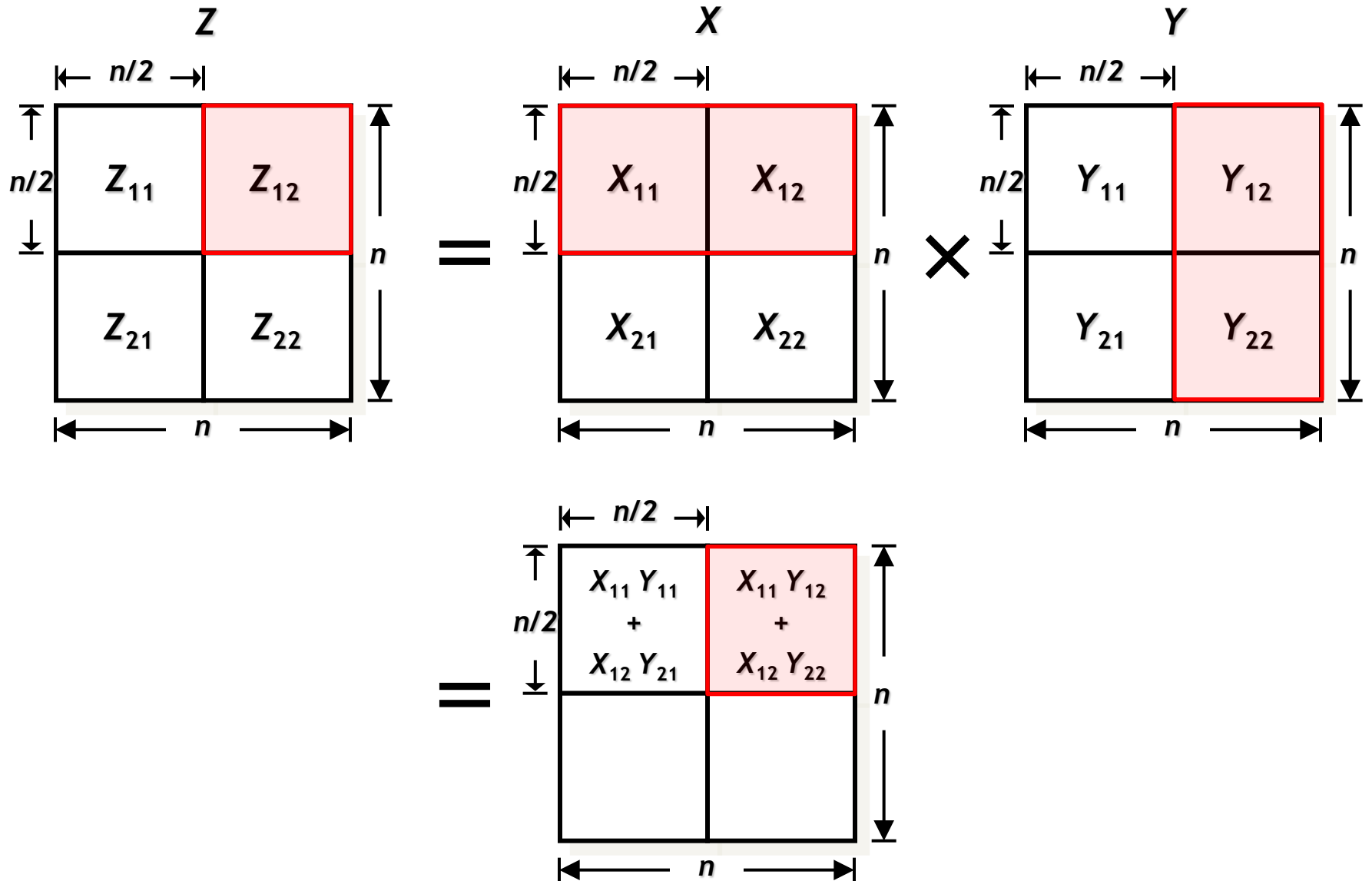
Recursive (Divide & Conquer) Matrix Multiplication



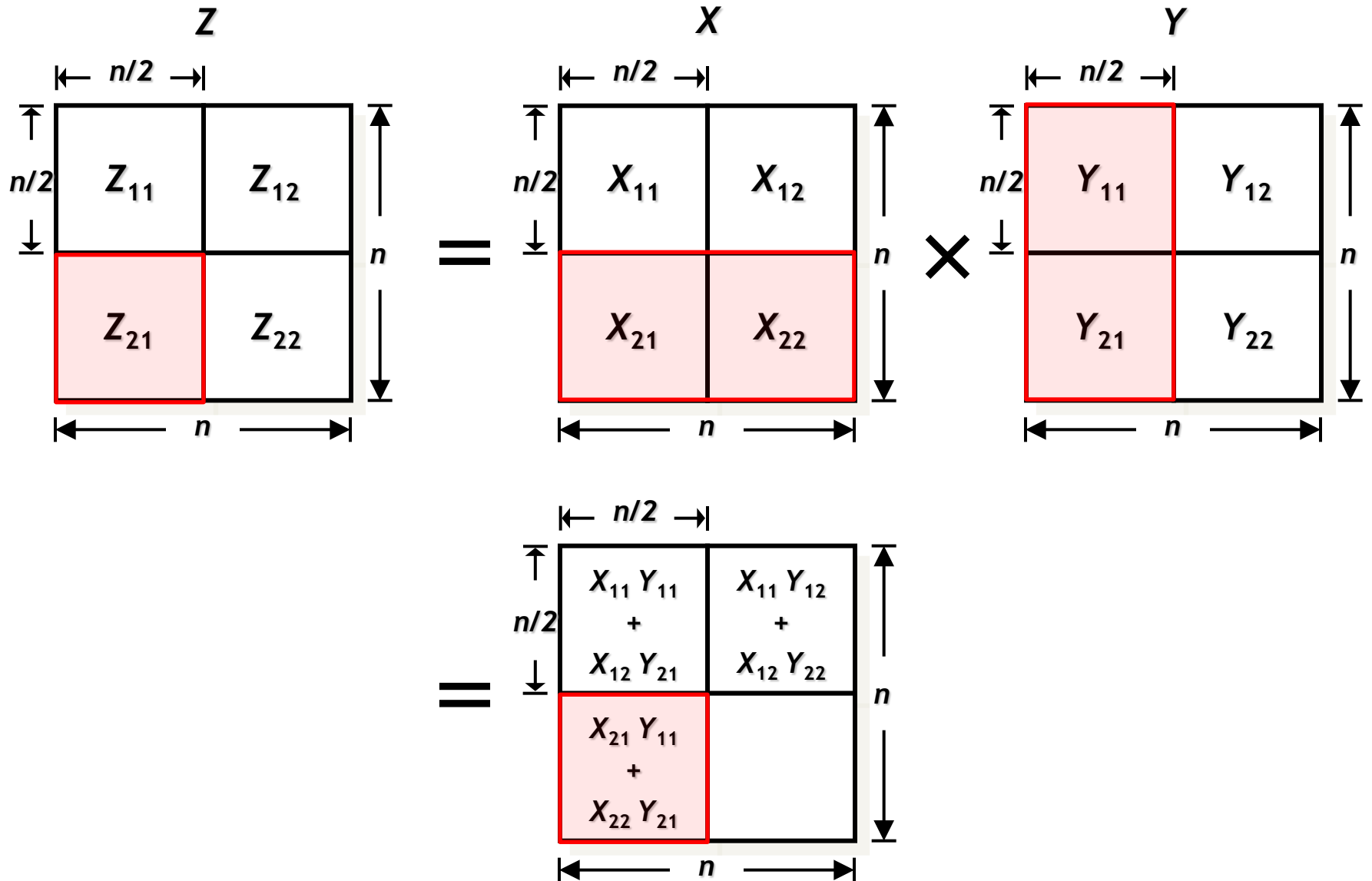
Recursive (Divide & Conquer) Matrix Multiplication



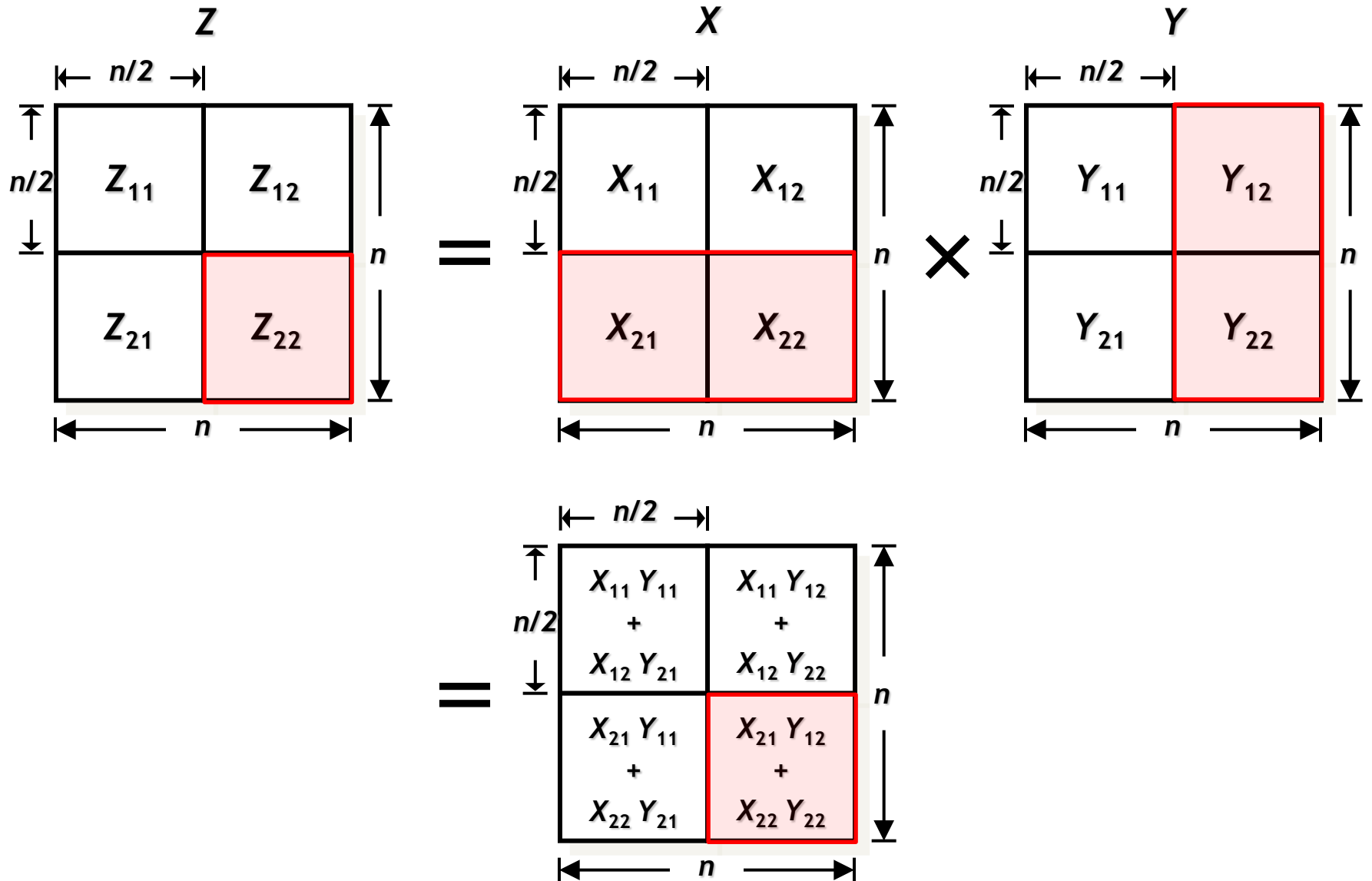
Recursive (Divide & Conquer) Matrix Multiplication



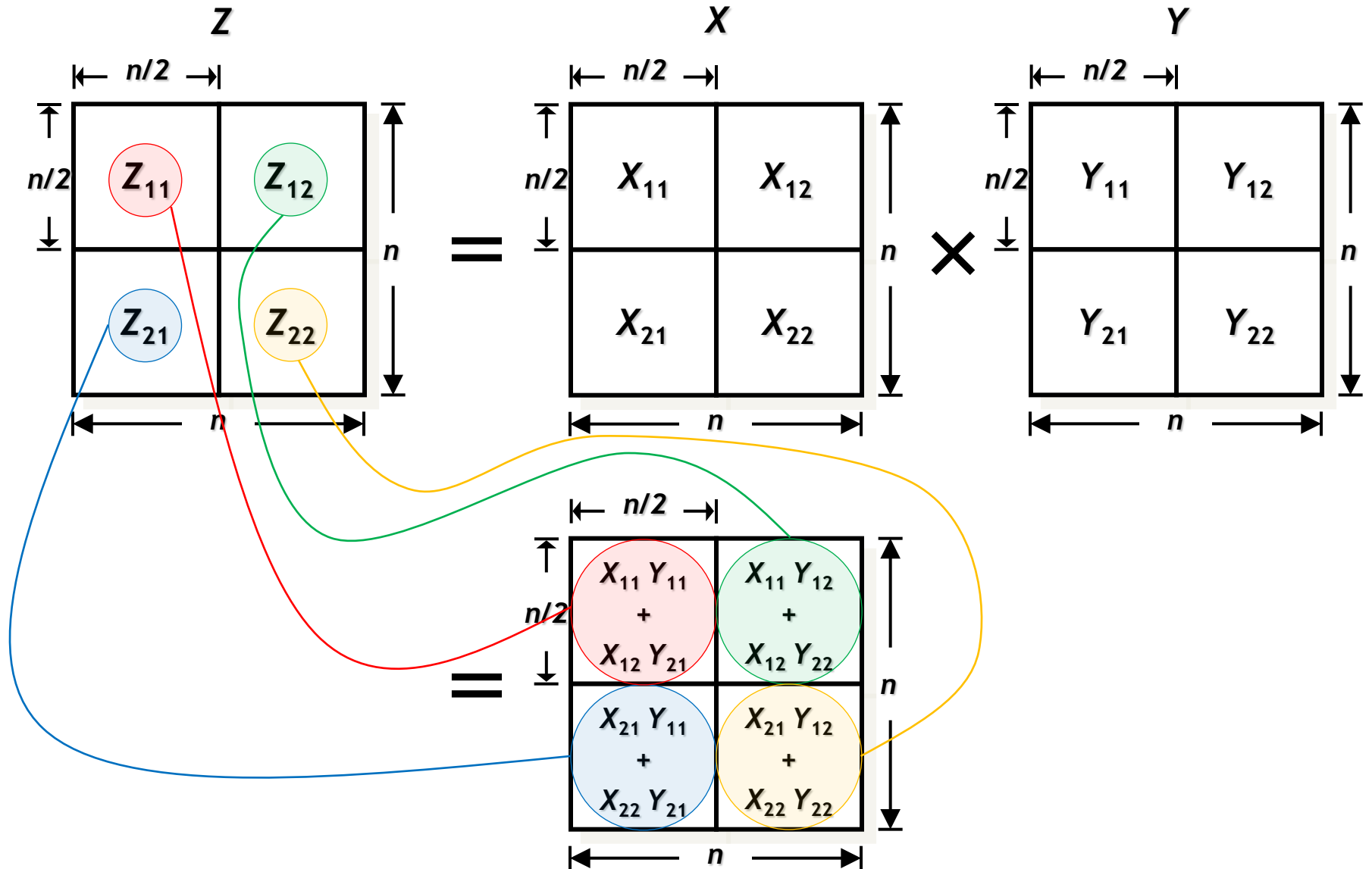
Recursive (Divide & Conquer) Matrix Multiplication



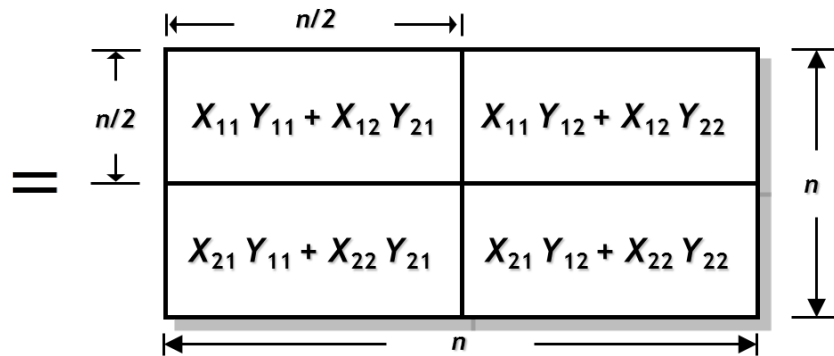
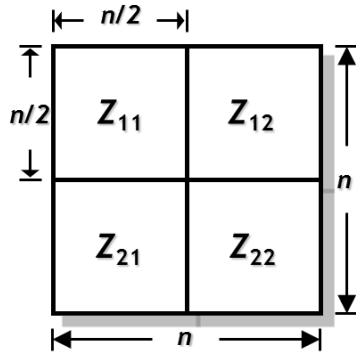
Recursive (Divide & Conquer) Matrix Multiplication



Recursive (Divide & Conquer) Matrix Multiplication



Recursive (Divide & Conquer) Matrix Multiplication



Rec-MM (X, Y) { X and Y are $n \times n$ matrices,
where $n = 2^k$ for integer $k \geq 0$ }

1. *Let* Z *be a new* $n \times n$ *matrix*
2. *if* $n = 1$ *then*
3. $Z \leftarrow X \cdot Y$
4. *else*
5. $Z_{11} \leftarrow \text{Rec-MM} (X_{11}, Y_{11}) + \text{Rec-MM} (X_{12}, Y_{21})$
6. $Z_{12} \leftarrow \text{Rec-MM} (X_{11}, Y_{12}) + \text{Rec-MM} (X_{12}, Y_{22})$
7. $Z_{21} \leftarrow \text{Rec-MM} (X_{21}, Y_{11}) + \text{Rec-MM} (X_{22}, Y_{21})$
8. $Z_{22} \leftarrow \text{Rec-MM} (X_{21}, Y_{12}) + \text{Rec-MM} (X_{22}, Y_{22})$
9. *endif*
10. *return* Z

recursive matrix products: 8

matrix sums: 4

$$T(n) = \begin{cases} \Theta(1), & \text{if } n = 1, \\ 8T\left(\frac{n}{2}\right) + \Theta(n^2), & \text{otherwise.} \end{cases}$$

$$= \Theta(n^3)$$

Strassen's Algorithms for Matrix Multiplication (MM)



In 1968 Volker Strassen came up with a recursive MM algorithm that runs asymptotically faster than the classical $\Theta(n^3)$ algorithm.

In each level of recursion the algorithm uses:

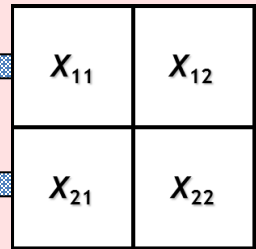
7 recursive matrix multiplications (instead of 8), and
18 matrix additions (instead of 4).

Strassen's MM: 10 Matrix Additions/Subtractions

row sums (4)

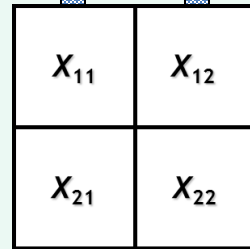
column diffs (4)

diagonal sums (2)



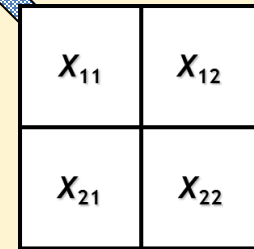
$$X_{r1} = x_{11} + x_{12}$$

$$X_{r2} = x_{21} + x_{22}$$

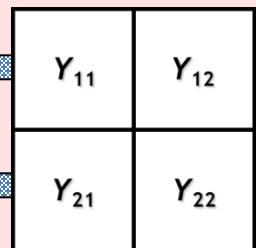


$$X_{c1} = x_{11} - x_{21}$$

$$X_{c2} = x_{12} - x_{22}$$

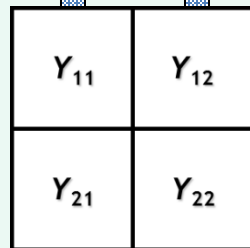


$$X_{d1} = x_{11} + x_{22}$$



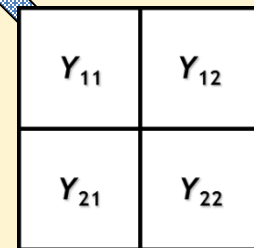
$$Y_{r1} = y_{11} + y_{12}$$

$$Y_{r2} = y_{21} + y_{22}$$



$$Y_{c1} = y_{11} - y_{21}$$

$$Y_{c2} = y_{12} - y_{22}$$

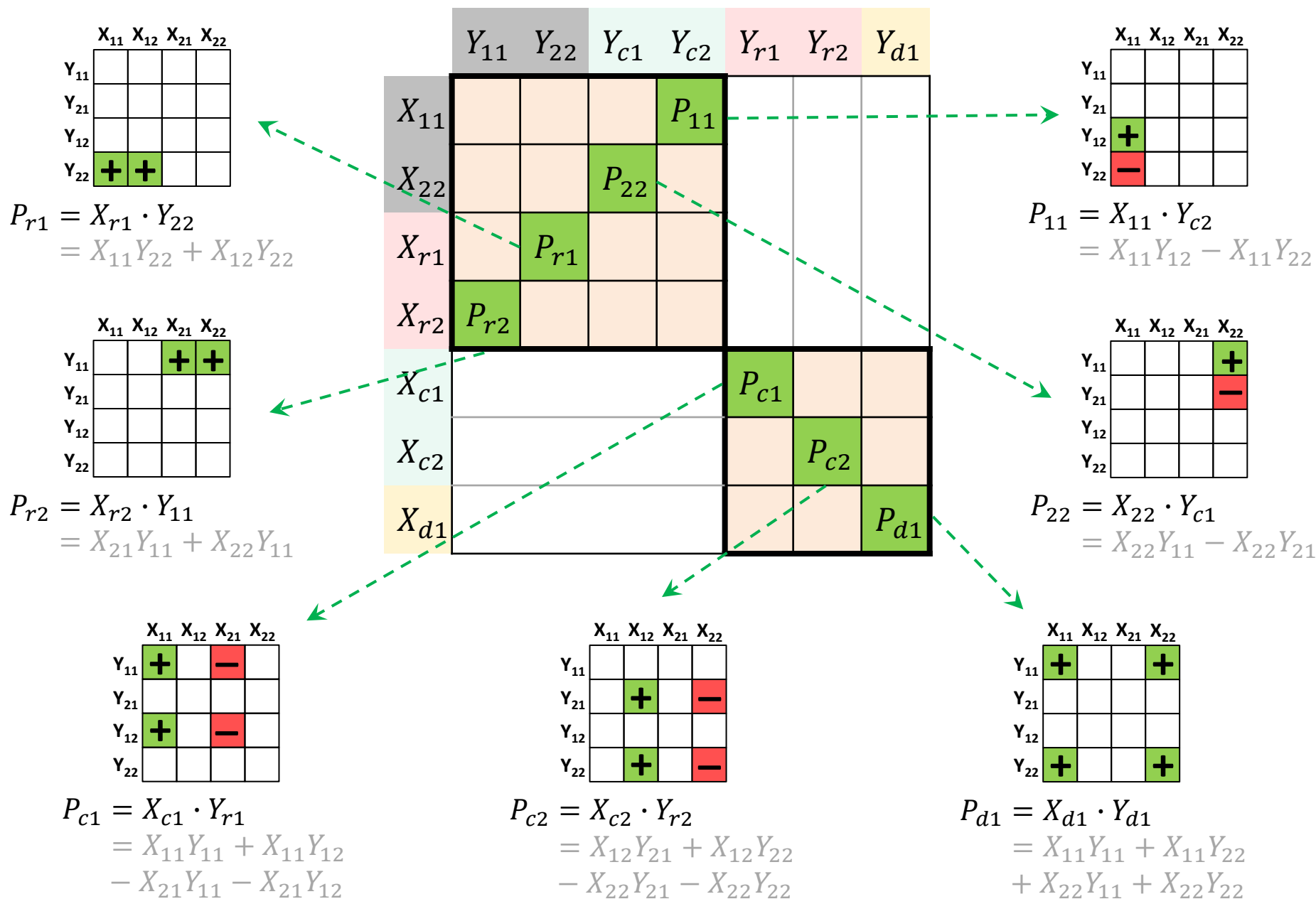


$$Y_{d1} = y_{11} + y_{22}$$

Strassen's MM: 7 Matrix Products

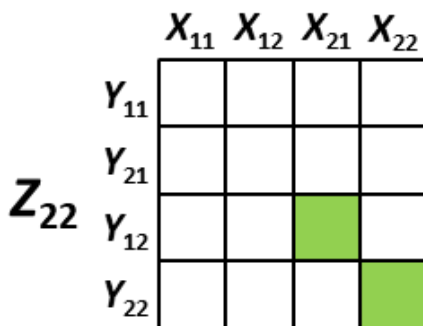
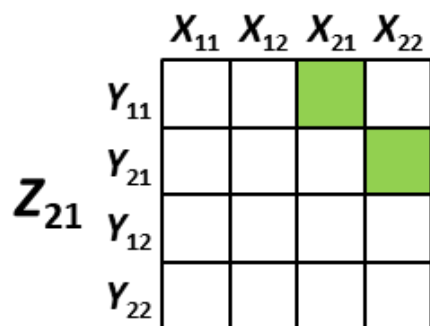
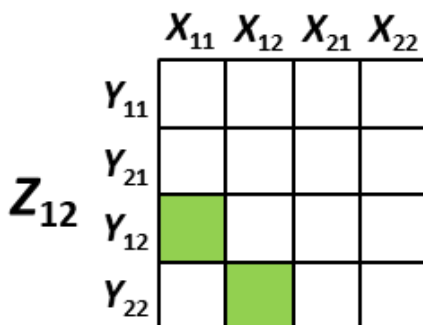
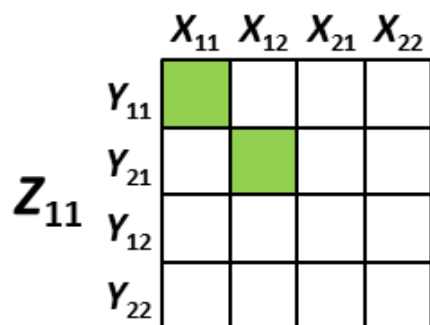
	Y_{11}	Y_{22}	Y_{c1}	Y_{c2}	Y_{r1}	Y_{r2}	Y_{d1}
X_{11}				P_{11}			
X_{22}			P_{22}				
X_{r1}		P_{r1}					
X_{r2}	P_{r2}						
X_{c1}					P_{c1}		
X_{c2}						P_{c2}	
X_{d1}							P_{d1}

Strassen's MM: 7 Matrix Products



Strassen's Matrix Multiplication

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{bmatrix} \times \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} X_{11}Y_{11} + X_{12}Y_{21} & X_{11}Y_{12} + X_{12}Y_{22} \\ X_{21}Y_{11} + X_{22}Y_{21} & X_{21}Y_{12} + X_{22}Y_{22} \end{bmatrix}$$



Strassen's Matrix Multiplication

	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & & & \\ Y_{12} & + & & & \\ Y_{22} & - & & & \end{matrix}$ P_{11}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & + \\ Y_{21} & & & & - \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$ P_{22}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & & & \\ Y_{12} & & & & \\ Y_{22} & + & + & & \end{matrix}$ P_{r1}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & + & + & \\ Y_{21} & & & & \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$ P_{r2}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & + & & - & \\ Y_{21} & & & & \\ Y_{12} & + & & - & \\ Y_{22} & & & & \end{matrix}$ P_{c1}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & + & & - \\ Y_{12} & & & & \\ Y_{22} & & + & & - \end{matrix}$ P_{c2}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & + & & & + \\ Y_{21} & & & & \\ Y_{12} & & & & \\ Y_{22} & + & & & + \end{matrix}$ P_{d1}
Z_{11}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & + & & & \\ Y_{21} & & + & & \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$						
Z_{12}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & & & \\ Y_{12} & + & & & \\ Y_{22} & & + & & \end{matrix}$						
Z_{21}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & + & \\ Y_{21} & & & & + \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$						
Z_{22}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & & & \\ Y_{12} & & & + & \\ Y_{22} & & & & + \end{matrix}$						

Strassen's Matrix Multiplication

	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & & & \\ Y_{12} & + & & & \\ Y_{22} & - & & & \end{matrix}$ P_{11}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & + \\ Y_{21} & & & & - \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$ P_{22}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & & & \\ Y_{12} & & & & \\ Y_{22} & + & + & & \end{matrix}$ P_{r1}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & + & + & \\ Y_{21} & & & & \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$ P_{r2}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & + & & - & \\ Y_{21} & & & & \\ Y_{12} & + & & - & \\ Y_{22} & & & & \end{matrix}$ P_{c1}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & + & & - \\ Y_{12} & & & & \\ Y_{22} & & + & & - \end{matrix}$ P_{c2}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & + & & & + \\ Y_{21} & & & & \\ Y_{12} & & & & \\ Y_{22} & + & & & + \end{matrix}$ P_{d1}
Z_{11}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & \boxed{} & & & \\ Y_{21} & & \boxed{} & & \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$						
Z_{12}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & & & \\ Y_{12} & \boxed{} & & & \\ Y_{22} & & \boxed{} & & \end{matrix}$						
Z_{21}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & \boxed{} & \\ Y_{21} & & & & \boxed{} \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$						
Z_{22}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & & & \\ Y_{12} & & & \boxed{} & \\ Y_{22} & & & \boxed{} & \boxed{} \end{matrix}$						

Strassen's Matrix Multiplication

	$\begin{matrix} X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & \\ Y_{21} & & & \\ Y_{12} & + & & \\ Y_{22} & - & & \end{matrix}$ P_{11}	$\begin{matrix} X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & + \\ Y_{21} & & & - \\ Y_{12} & & & \\ Y_{22} & & & \end{matrix}$ P_{22}	$\begin{matrix} X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & \\ Y_{21} & & & \\ Y_{12} & & & \\ Y_{22} & + & + & \end{matrix}$ P_{r1}	$\begin{matrix} X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & + & + \\ Y_{21} & & & \\ Y_{12} & & & \\ Y_{22} & & & \end{matrix}$ P_{r2}	$\begin{matrix} X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & + & & - \\ Y_{21} & & & \\ Y_{12} & + & & - \\ Y_{22} & & & \end{matrix}$ P_{c1}	$\begin{matrix} X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & \\ Y_{21} & + & & - \\ Y_{12} & & & \\ Y_{22} & + & & - \end{matrix}$ P_{c2}	$\begin{matrix} X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & + & & + \\ Y_{21} & & & \\ Y_{12} & & & \\ Y_{22} & + & & + \end{matrix}$ P_{d1}
Z_{11} $\begin{matrix} X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & + & & + \\ Y_{21} & & + & \\ Y_{12} & & & \\ Y_{22} & + & & + \end{matrix}$							$\begin{matrix} & + & & \\ + & & & + \\ & & + & \\ & & & \\ + & & & + \end{matrix}$
Z_{12} $\begin{matrix} X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & \\ Y_{21} & & & \\ Y_{12} & + & & \\ Y_{22} & & + & \end{matrix}$							
Z_{21} $\begin{matrix} X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & + \\ Y_{21} & & & + \\ Y_{12} & & & \\ Y_{22} & & & \end{matrix}$							
Z_{22} $\begin{matrix} X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & \\ Y_{21} & & & \\ Y_{12} & & + & \\ Y_{22} & & + & + \end{matrix}$							

Strassen's Matrix Multiplication

The diagram illustrates the construction of a 4x4 matrix P from 2x2 blocks P_{ij} . The blocks are arranged in a 4x4 grid with dashed lines separating them. The blocks are labeled P_{11} , P_{22} , P_{r1} , P_{r2} , P_{c1} , P_{c2} , and P_{d1} . The blocks P_{c1} and P_{c2} are highlighted with green and red colors.

The blocks are defined as follows:

- P_{11} : $\begin{bmatrix} + & - \\ - & + \end{bmatrix}$
- P_{22} : $\begin{bmatrix} + & - \\ - & + \end{bmatrix}$
- P_{r1} : $\begin{bmatrix} + & + \\ + & + \end{bmatrix}$
- P_{r2} : $\begin{bmatrix} + & + \\ + & + \end{bmatrix}$
- P_{c1} : $\begin{bmatrix} + & - \\ - & + \end{bmatrix}$
- P_{c2} : $\begin{bmatrix} + & - \\ - & + \end{bmatrix}$
- P_{d1} : $\begin{bmatrix} + & + \\ + & + \end{bmatrix}$

Strassen's Matrix Multiplication

	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & & & \\ Y_{12} & + & & & \\ Y_{22} & - & & & \end{matrix}$ P_{11}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & + \\ Y_{21} & & & & - \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$ P_{22}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & & & \\ Y_{12} & & & & \\ Y_{22} & + & + & & \end{matrix}$ P_{r1}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & + & + & \\ Y_{21} & & & & \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$ P_{r2}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & + & & - & \\ Y_{21} & + & & - & \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$ P_{c1}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & + & & - & \\ Y_{12} & & & & \\ Y_{22} & + & & - & \end{matrix}$ P_{c2}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & + & & & + \\ Y_{21} & & & & \\ Y_{12} & & & & \\ Y_{22} & + & & & + \end{matrix}$ P_{d1}
Z_{11}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & + & & & + \\ Y_{21} & & + & & - \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$		$\begin{matrix} & & & & - \\ & & & & \\ & & & & \\ & & & & \\ - & - & & & \end{matrix}$			$\begin{matrix} & & & & + \\ & & & & \\ & & + & & - \\ & & & & \\ & + & & & - \end{matrix}$	$\begin{matrix} & & & & + \\ & & & & \\ & + & & & + \\ & & & & \\ & + & & & + \end{matrix}$
Z_{12}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & & & \\ Y_{12} & + & & & \\ Y_{22} & & + & & \end{matrix}$						
Z_{21}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & + & \\ Y_{21} & & & & + \\ Y_{12} & & & & \\ Y_{22} & & & & \end{matrix}$						
Z_{22}	$\begin{matrix} & X_{11} & X_{12} & X_{21} & X_{22} \\ Y_{11} & & & & \\ Y_{21} & & & & \\ Y_{12} & & & + & \\ Y_{22} & & & + & + \end{matrix}$						

Strassen's Matrix Multiplication

	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} + Y_{22} - P_{11}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} P_{22}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} + + P_{r1}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} P_{r2}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} + Y_{21} Y_{12} + Y_{22} P_{c1}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} + Y_{12} Y_{22} + P_{c2}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} + Y_{21} Y_{12} Y_{22} + P_{d1}
Z_{11}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} + - Z_{11}	-	-			+	+
Z_{12}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} Z_{12}						
Z_{21}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} Z_{21}						
Z_{22}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} Z_{22}						

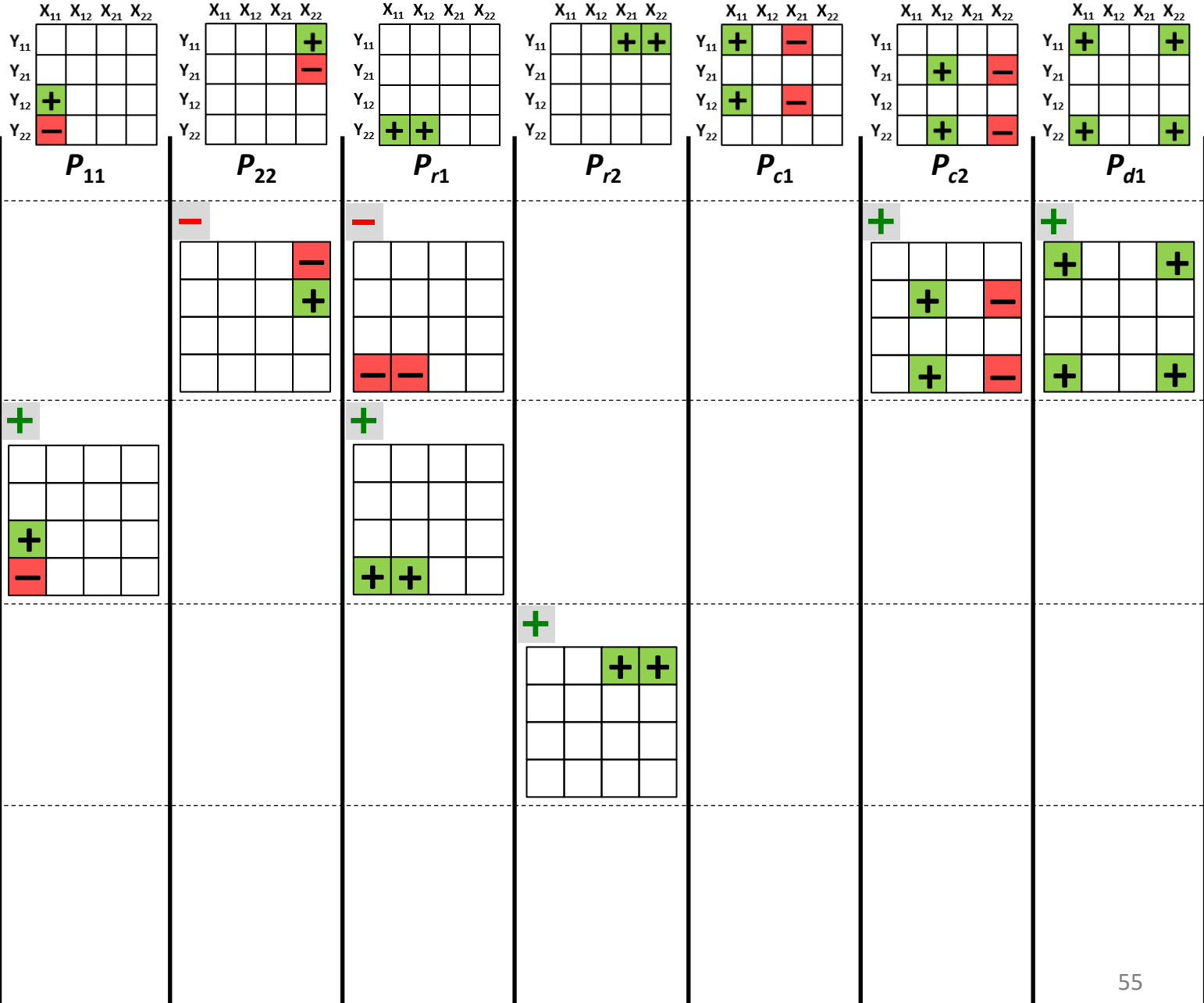
Strassen's Matrix Multiplication

	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} + Y_{22} - P_{11}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} P_{22}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} + + P_{r1}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} P_{r2}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} + Y_{21} Y_{12} + Y_{22} P_{c1}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} + Y_{12} Y_{22} + P_{c2}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} + Y_{21} Y_{12} Y_{22} + P_{d1}
Z_{11}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} Z_{11}	-	-			+	+
Z_{12}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} Z_{12}			+			
Z_{21}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} Z_{21}						
Z_{22}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} Z_{22}						

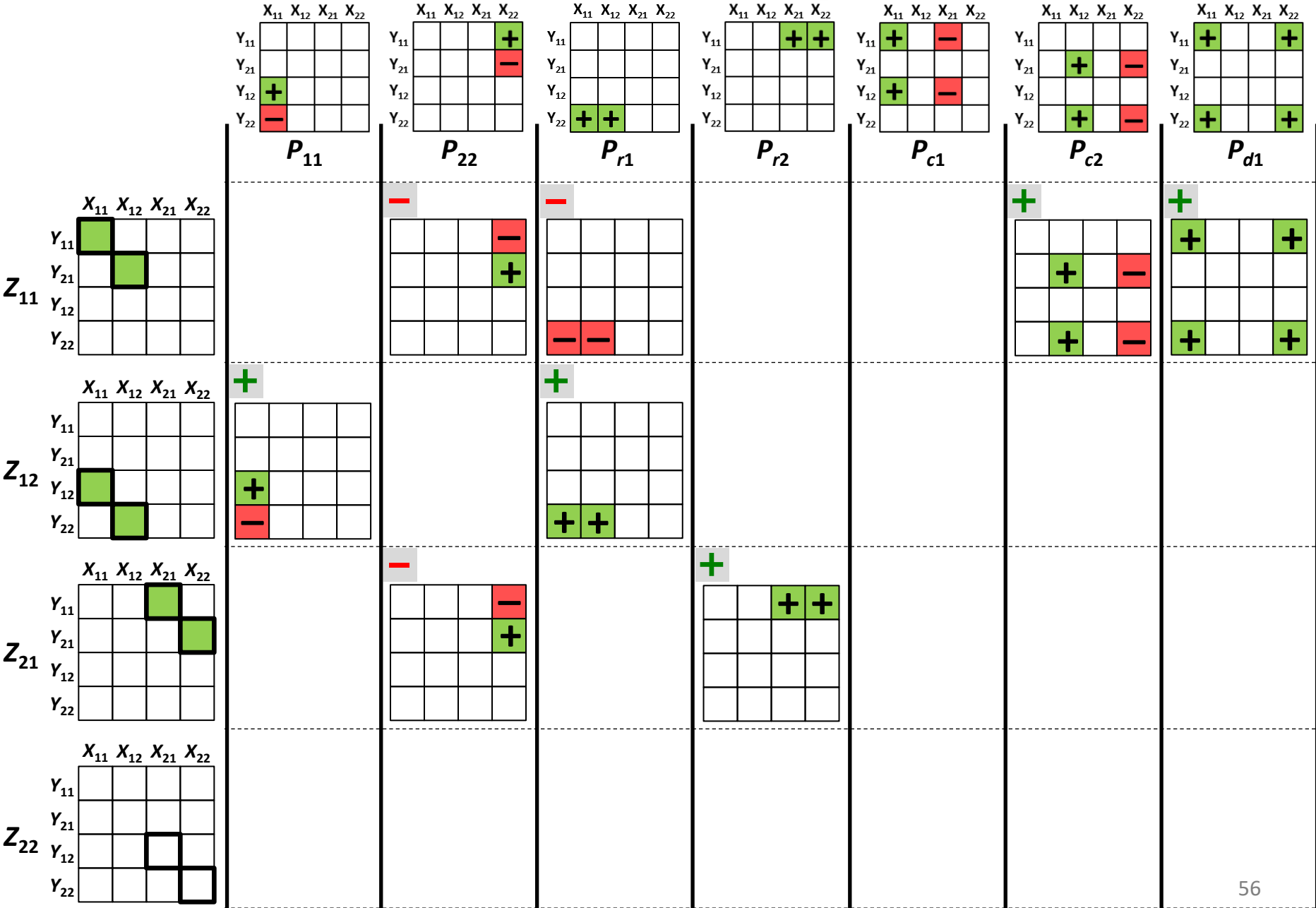
Strassen's Matrix Multiplication

	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} + Y_{22} - P_{11}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} P_{22}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} + + P_{r1}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} Y_{12} Y_{22} P_{r2}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} + Y_{21} Y_{12} + Y_{22} P_{c1}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} Y_{21} + Y_{12} Y_{22} + P_{c2}	$X_{11} \ X_{12} \ X_{21} \ X_{22}$ Y_{11} + Y_{21} Y_{12} Y_{22} + P_{d1}
Z_{11}		-	-			+	+
Z_{12}	+		+				
Z_{21}							
Z_{22}							

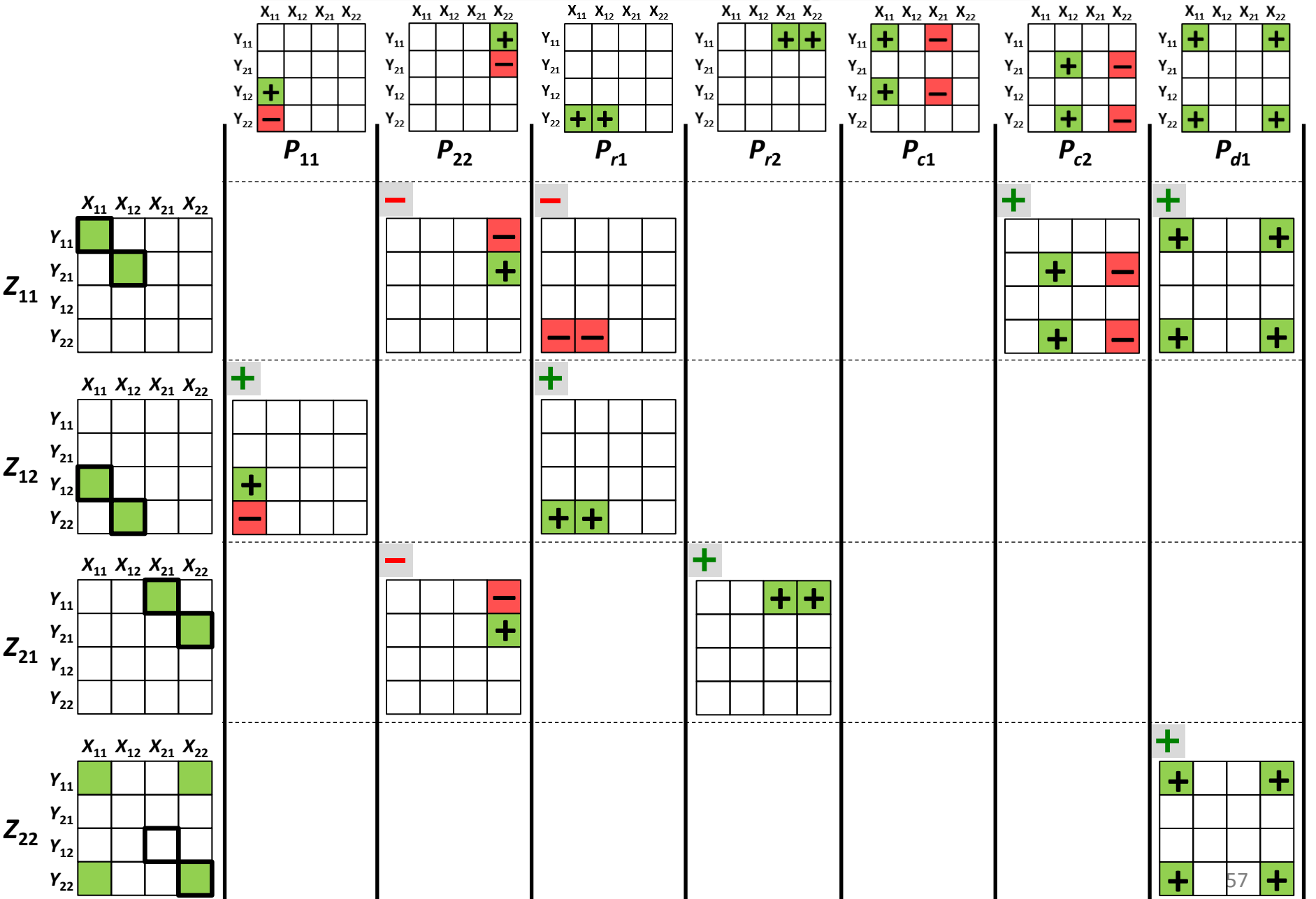
Strassen's Matrix Multiplication



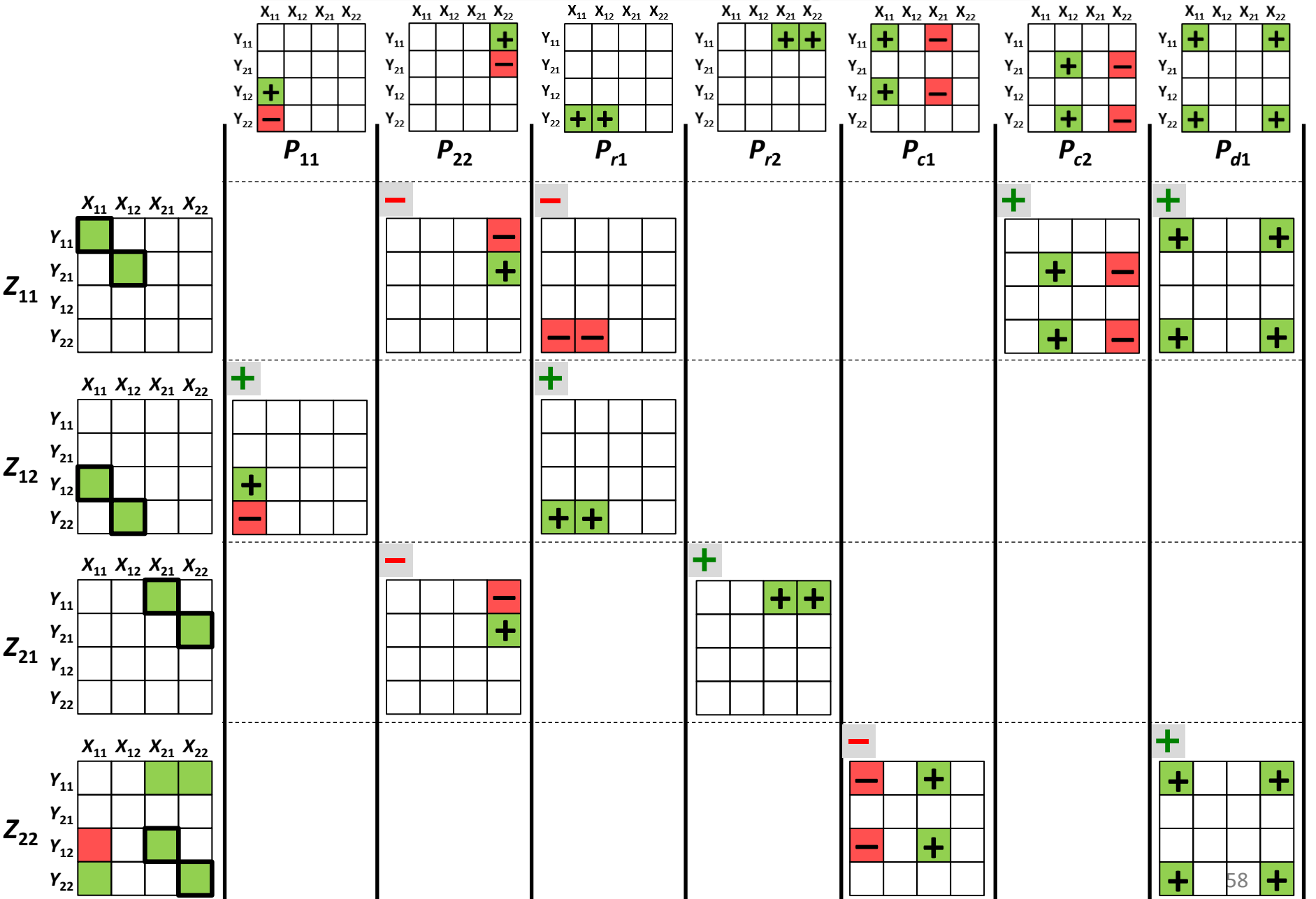
Strassen's Matrix Multiplication



Strassen's Matrix Multiplication



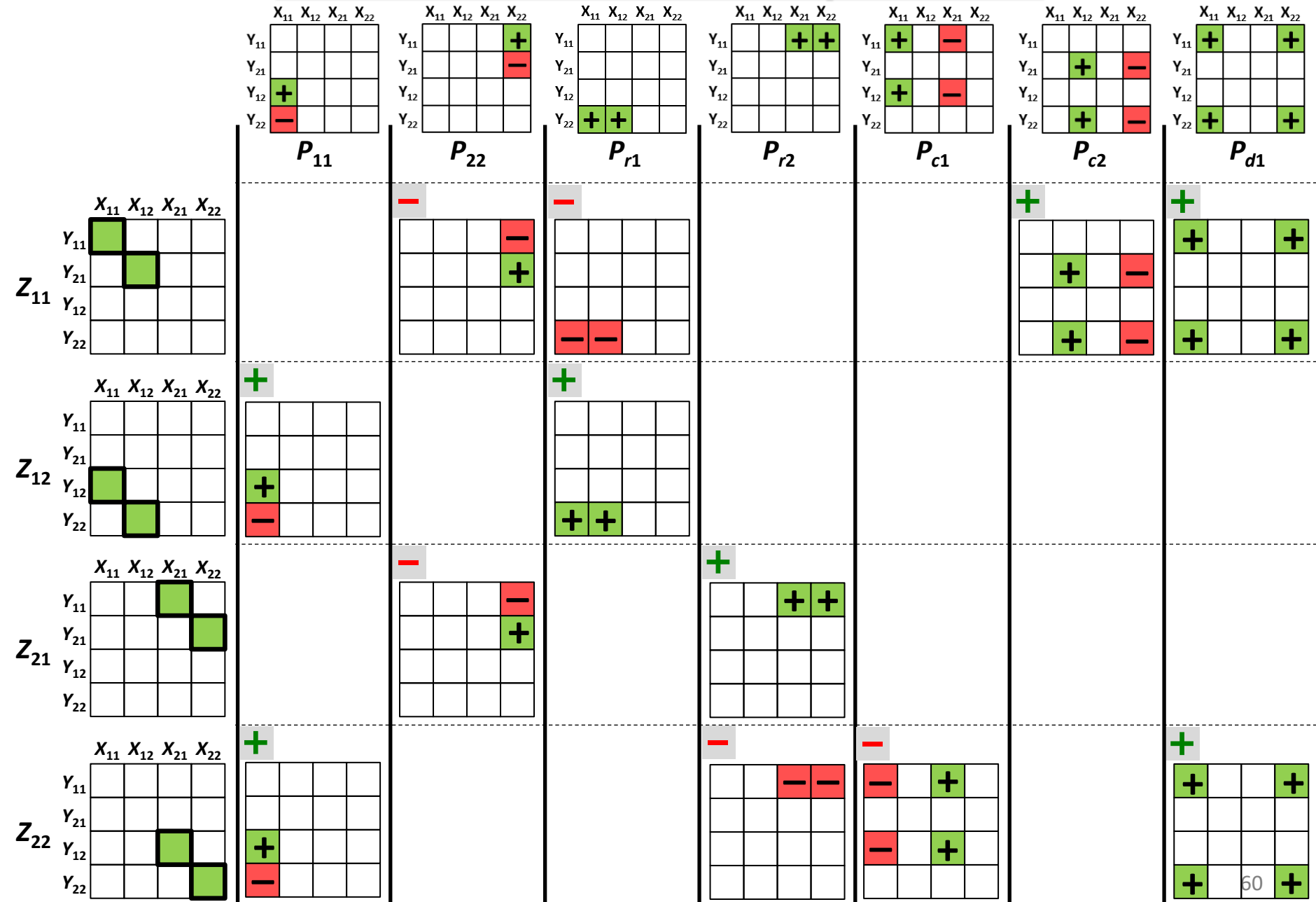
Strassen's Matrix Multiplication



Strassen's Matrix Multiplication

	$X_{11} \ X_{12} \ X_{21} \ X_{22}$	$X_{11} \ X_{12} \ X_{21} \ X_{22}$	$X_{11} \ X_{12} \ X_{21} \ X_{22}$	$X_{11} \ X_{12} \ X_{21} \ X_{22}$	$X_{11} \ X_{12} \ X_{21} \ X_{22}$	$X_{11} \ X_{12} \ X_{21} \ X_{22}$	$X_{11} \ X_{12} \ X_{21} \ X_{22}$
Y_{11}							
Y_{21}							
Y_{12}							
Y_{22}							
	P_{11}	P_{22}	P_{r1}	P_{r2}	P_{c1}	P_{c2}	P_{d1}
Z_{11}							
Z_{12}							
Z_{21}							
Z_{22}							

Strassen's Matrix Multiplication



Strassen's Matrix Multiplication

$$\begin{array}{|c|c|} \hline Z_{11} & Z_{12} \\ \hline Z_{21} & Z_{22} \\ \hline \end{array} = \begin{array}{|c|c|} \hline X_{11} & X_{12} \\ \hline X_{21} & X_{22} \\ \hline \end{array} \times \begin{array}{|c|c|} \hline Y_{11} & Y_{12} \\ \hline Y_{21} & Y_{22} \\ \hline \end{array} = \begin{array}{|c|c|} \hline X_{11}Y_{11} + X_{12}Y_{21} & X_{11}Y_{12} + X_{12}Y_{22} \\ \hline X_{21}Y_{11} + X_{22}Y_{21} & X_{21}Y_{12} + X_{22}Y_{22} \\ \hline \end{array}$$

Sums:

$$\begin{array}{ll}
 X_{r1} = X_{11} + X_{12} & Y_{r1} = Y_{11} + Y_{12} \\
 X_{r2} = X_{21} + X_{22} & Y_{r2} = Y_{21} + Y_{22} \\
 X_{c1} = X_{11} - X_{21} & Y_{c1} = Y_{11} - Y_{21} \\
 X_{c2} = X_{12} - X_{22} & Y_{c2} = Y_{12} - Y_{22} \\
 X_{d1} = X_{11} + X_{22} & Y_{d1} = Y_{11} + Y_{22}
 \end{array}$$

$$\begin{array}{|c|c|c|c|} \hline -P_{r1} & & -P_{22} & +P_{r1} \\ \hline +P_{d1} & & +P_{c2} & \\ \hline +P_{r2} & & -P_{22} & -P_{r2} \\ \hline & & +P_{d1} & -P_{c1} \\ \hline \end{array}$$

Products:

$$\begin{array}{ll}
 P_{11} = X_{11} \cdot Y_{c2} & P_{c1} = X_{c1} \cdot Y_{r1} \\
 P_{22} = X_{22} \cdot Y_{c1} & P_{c2} = X_{c2} \cdot Y_{r2} \\
 P_{r1} = X_{r1} \cdot Y_{22} & P_{d1} = X_{d1} \cdot Y_{d1} \\
 P_{r2} = X_{r2} \cdot Y_{11} &
 \end{array}$$

Running Time:

$$T(n) = \begin{cases} \Theta(1), & \text{if } n = 1, \\ 7T\left(\frac{n}{2}\right) + \Theta(n^2), & \text{otherwise.} \end{cases}$$

$$= \Theta(n^{\log_2 7}) = O(n^{2.81})$$

Deriving Strassen's Algorithm

Use the *Feynman Algorithm*:

Step 1: write down the problem

Step 2: think real hard

Step 3: write down the solution

Deriving Strassen's Algorithm

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \Rightarrow \underbrace{\begin{bmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix}}_X \underbrace{\begin{bmatrix} e \\ g \\ f \\ h \end{bmatrix}}_Y = \underbrace{\begin{bmatrix} p \\ r \\ q \\ s \end{bmatrix}}_Z$$

We will try to minimize the number of multiplications needed to evaluate Z using special matrix products that are easy to compute.

<u>Type</u>	<u>Product</u>	<u>#Mults</u>
($\cdot$)	$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e \\ g \end{bmatrix} = \begin{bmatrix} ae + bg \\ ce + dg \end{bmatrix}$	4
(A)	$\begin{bmatrix} a & a \\ a & a \end{bmatrix} \begin{bmatrix} e \\ g \end{bmatrix} = \begin{bmatrix} a(e + g) \\ a(e + g) \end{bmatrix}$	1
(B)	$\begin{bmatrix} a & a \\ -a & -a \end{bmatrix} \begin{bmatrix} e \\ g \end{bmatrix} = \begin{bmatrix} a(e + g) \\ -a(e + g) \end{bmatrix}$	1
(C)	$\begin{bmatrix} a & 0 \\ a - b & b \end{bmatrix} \begin{bmatrix} e \\ g \end{bmatrix} = \begin{bmatrix} ae \\ ae + b(g - e) \end{bmatrix}$	2
(D)	$\begin{bmatrix} a & b - a \\ 0 & b \end{bmatrix} \begin{bmatrix} e \\ g \end{bmatrix} = \begin{bmatrix} a(e - g) + bg \\ bg \end{bmatrix}$	2

Deriving Strassen's Algorithm

$$\begin{bmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix}$$

Deriving Strassen's Algorithm

$$\begin{bmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix} = \underbrace{\begin{bmatrix} b & b & 0 & 0 \\ b & b & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type A (1 Mult)}} + \begin{bmatrix} a-b & 0 & 0 & 0 \\ c-b & d-b & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix}$$

Deriving Strassen's Algorithm

$$\begin{aligned}
 \begin{bmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix} &= \underbrace{\begin{bmatrix} b & b & 0 & 0 \\ b & b & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type A (1 Mult)}} + \begin{bmatrix} a-b & 0 & 0 & 0 \\ c-b & d-b & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix} \\
 &= \underbrace{\begin{bmatrix} b & b & 0 & 0 \\ b & b & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type A (1 Mult)}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & c & c \\ 0 & 0 & c & c \end{bmatrix}}_{\text{Type A (1 Mult)}} + \begin{bmatrix} a-b & 0 & 0 & 0 \\ c-b & d-b & 0 & 0 \\ 0 & 0 & a-c & b-c \\ 0 & 0 & 0 & d-c \end{bmatrix}
 \end{aligned}$$

Deriving Strassen's Algorithm

$$\begin{aligned}
 \begin{bmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix} &= \underbrace{\begin{bmatrix} b & b & 0 & 0 \\ b & b & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type A (1 Mult)}} + \begin{bmatrix} a-b & 0 & 0 & 0 \\ c-b & d-b & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & 0 & c & d \end{bmatrix} \\
 &= \underbrace{\begin{bmatrix} b & b & 0 & 0 \\ b & b & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type A (1 Mult)}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & c & c \\ 0 & 0 & c & c \end{bmatrix}}_{\text{Type A (1 Mult)}} + \begin{bmatrix} a-b & 0 & 0 & 0 \\ c-b & d-b & 0 & 0 \\ 0 & 0 & a-c & b-c \\ 0 & 0 & 0 & d-c \end{bmatrix} \\
 &= \underbrace{\begin{bmatrix} b & b & 0 & 0 \\ b & b & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type A (1 Mult)}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & c & c \\ 0 & 0 & c & c \end{bmatrix}}_{\text{Type A (1 Mult)}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ c-b & 0 & 0 & c-b \\ -(c-b) & 0 & 0 & -(c-b) \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type B (1 Mult)}} + \underbrace{\begin{bmatrix} a-b & 0 & 0 & 0 \\ 0 & d-b & 0 & b-c \\ c-b & 0 & a-c & 0 \\ 0 & 0 & 0 & d-c \end{bmatrix}}_{\text{Type B (1 Mult)}}
 \end{aligned}$$

Deriving Strassen's Algorithm

$$\begin{aligned}
 \begin{bmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix} &= \underbrace{\begin{bmatrix} b & b & 0 & 0 \\ b & b & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type A (1 Mult)}} + \begin{bmatrix} a-b & 0 & 0 & 0 \\ c-b & d-b & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix} \\
 &= \underbrace{\begin{bmatrix} b & b & 0 & 0 \\ b & b & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type A (1 Mult)}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & c & c \\ 0 & 0 & c & c \end{bmatrix}}_{\text{Type A (1 Mult)}} + \begin{bmatrix} a-b & 0 & 0 & 0 \\ c-b & d-b & 0 & 0 \\ 0 & 0 & a-c & b-c \\ 0 & 0 & 0 & d-c \end{bmatrix} \\
 &= \underbrace{\begin{bmatrix} b & b & 0 & 0 \\ b & b & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type A (1 Mult)}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & c & c \\ 0 & 0 & c & c \end{bmatrix}}_{\text{Type A (1 Mult)}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ c-b & 0 & 0 & c-b \\ -(c-b) & 0 & 0 & -(c-b) \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type B (1 Mult)}} + \begin{bmatrix} a-b & 0 & 0 & 0 \\ 0 & d-b & 0 & b-c \\ c-b & 0 & a-c & 0 \\ 0 & 0 & 0 & d-c \end{bmatrix} \\
 &= \underbrace{\begin{bmatrix} b & b & 0 & 0 \\ b & b & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type A (1 Mult)}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & c & c \\ 0 & 0 & c & c \end{bmatrix}}_{\text{Type A (1 Mult)}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ c-b & 0 & 0 & c-b \\ -(c-b) & 0 & 0 & -(c-b) \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type B (1 Mult)}} + \underbrace{\begin{bmatrix} a-b & 0 & 0 & 0 \\ 0 & d-b & 0 & b-c \\ (a-b)-(a-c) & 0 & a-c & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Type C (2 Mult)}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & d-b & 0 & (d-c)-(d-b) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & d-c \end{bmatrix}}_{\text{Type D (2 Mult)}}
 \end{aligned}$$

Algorithms for Multiplying Two $n \times n$ Matrices

A recursive algorithm based on multiplying two $m \times m$ matrices using k multiplications will yield an $O(n^{\log_m k})$ algorithm.

To beat Strassen's algorithm: $\log_m k < \log_2 7 \Rightarrow k < m^{\log_2 7}$.

So, for a 3×3 matrix, we must have: $k < 3^{\log_2 7} < 22$.

But the best known algorithm uses 23 multiplications!

Inventor	Year	Complexity
Classical	—	$\Theta(n^3)$
Volker Strassen	1968	$\Theta(n^{2.807})$
Victor Pan (multiply two 70×70 matrices using 143,640 multiplications)	1978	$\Theta(n^{2.795})$
Don Coppersmith & Shmuel Winograd (arithmetic progressions)	1990	$\Theta(n^{2.3737})$
Andrew Stothers	2010	$\Theta(n^{2.3736})$
Virginia Williams	2011	$\Theta(n^{2.3727})$

Lower bound: $\Omega(n^2)$ (why?)

*When the math
gets tough*

*Analysis of Algorithms
in
Rhymes and Pictures*

ChatGPT 4o mini



Iterative Matrix Multiplication: A Rhyming Guide

When X and Y are matrices two,
And Z is the product, fresh and new,
We multiply them step by step,
Row by column—the process is kept!

Step 1: From X , take the very first row,
And from Y , the first column and go.
Multiply in pairs, then sum them up,
The first entry of Z fills the cup!

Step 2: Move to the next, row by row,
And match with columns as they flow.
Each product fills Z , slot by slot,
Till all positions are properly caught.

Step 3: Repeat the steps until it's done,
All rows and columns, everyone.
Now Z is filled from left to right,
A matrix formed, clean and bright!



Complexity in Rhyme

Matrix multiplication's iterative way,
Is $\Theta(n^3)$ —that's what they say.
For each row, column, and every dot,
The work multiplies—a busy plot.

Though faster methods do exist,
This basic approach gets the gist.
For simple tasks, it does the deed,
And fills your matrix with what you need!





Recursive Matrix Multiplication: A Rhyming Tale

When X and Y are large, you see,
Recursively break and make easy.
Instead of rows and columns straight,
We split and conquer—watch the fate!

Step 1: Split X and Y into four,
Smaller blocks, that's what they're for!
Each quadrant takes its turn to play,
In smaller steps, they pave the way.

Step 2: Recursively, it calls each part,
Multiplying blocks, smart from the start.
Then sums them up, combining neat,
Till Z , the product, is complete.

Step 3: Each block in Z forms its piece,
From all the recursive work's release.
When all the steps are fully done,
The whole matrix is joined as one!



Complexity in Rhyme

The recursive method, though it's slow,
At $O(n^3)$, it tends to grow.
But for big tasks, recursion's clear,
Breaking problems far and near.

Though not the fastest you might see,
It still divides things carefully.
For simpler tasks or learning ways,
Recursion's steps deserve some praise!





Strassen's Algorithm: A Rhyming Tale

When matrices grow large and tall,
And multiplying them takes a haul,
Strassen's method comes to play,
To speed things up and save the day!

Step 1: Split X and Y in four,
Submatrices, just like before.
But Strassen's trick—here's what's new:
Seven products, not eight to do!

Step 2: Compute those products—just a few,
With smart combinations, here's the clue:
P1 through P7 you will find,
Each one's a shortcut, cleverly designed.

Step 3: Combine those products, add and blend,
To get the result in the end.
Each block of Z is formed just right,
Fewer steps, and less in sight!

Step 4: Recursively, it calls once more,
Till matrices are small, just like before.
And in the end, with fewer steps,
Strassen's method shines its best.



Complexity in Rhyme

Strassen's trick is fast and sweet,
 $\Theta(n^{\log_2 7})$ can't be beat!
By cutting down the steps to take,
It makes large products quick to make.

For matrices big and wide,
Strassen's method saves some time.
It multiplies with fewer strides,
And keeps your math work quick and prime!

